Quantitative Data Analysis and Reporting

What do we want to know? What data would answer it? Which method fits? What can the output support?

English lesson handout. Begin with data and descriptive statistics, then work through four detailed methods. Open the interactive lesson for everyday stories and the full study PDFs. Further methods remain linked from the map.

Open the interactive lesson

Learning aims

By the end of the lesson, you should be able to:

1. Report a sample clearly with appropriate descriptive statistics, units and denominators.

2. Explain the questions answered by t-tests, correlation, one-way ANOVA and chi-square, and read their main outputs.

3. Find how a real paper used and reported an analysis, then use the method map for further methods.

4. Consult the glossary and use JARS-Quant to check how analyses and results are reported.

Start with people, variables and values

Before choosing a test, say what one row means and what each column records.

Illustration of a data layout; these are fictional records.

Learner	Study group	Quiz score / 100	Preferred format
A	Quiet room	72	Video
B	Music room	81	Text
C	Quiet room	68	Text

- A case is the unit recorded in a row. Here it is one learner. A variable is a characteristic recorded for every learner.

- A score is a measured amount. A category is a label: room type or preferred format. Coding Video = 1 does not turn the label into an amount.

- Two rows from the same person are linked. Learners in the same class can also be linked. Check the design before treating rows as independent.

Pause and explain

Which column would you average, and which would you count?

Compare your explanation

Average the quiz scores. Count learners in each room or format category. Do not average the ID or category codes.

Textbook reading: Fein et al. (2022), §§1.1-1.2 (pp. 3-5; PDF pp. 11-13); Larson-Hall (2009), Chapters 2-3. Open the free Fein et al. textbook

Describe the sample before testing a claim

Readers first need to know how much data you have, what is typical and how much people differ.

What to report	What it tells the reader	How to choose
n and missing values	How many usable observations support this summary.	Give the valid n for each variable, not just the number recruited.
Counts and percentages	How common each category is.	State the denominator: 12 of 20 is different from 12 of 100.
Mean (M) and standard deviation (SD)	The average and how spread out individual scores are.	Read both together and inspect the distribution. SD is not the uncertainty of the mean.
Median and IQR	The middle value and the spread of the middle half.	Useful for a skewed distribution or extreme observations. IQR = Q3 − Q1.

A mean of 75 can hide tightly clustered scores or a very wide spread. A graph and SD help you tell these situations apart. Describing this sample does not establish that it represents every learner.

Pause and explain

Why is “M = 75” an incomplete description?

Compare your explanation

We still need n, the score scale, missingness and spread, and ideally a view of the distribution.

Textbook reading: Fein et al. (2022), §§1.1-1.2 (pp. 3-5; PDF pp. 11-13); Larson-Hall (2009), Chapters 2-3. Open the free Fein et al. textbook

Find a descriptive result in a real paper

Separate what students report believing from what a learning test demonstrates.

Nami surveyed 381 university students about language-learning apps. Table 1 reports survey-item means, SDs and response percentages. The labels below shorten the original item wording.

Table 1: selected reported values

Item topic	M	SD	Agree / strongly agree
Preference for free apps	4.10	0.896	76.4%
Perceived usefulness for vocabulary	4.18	0.749	86.3%

These are five-point survey responses. The paper summarises them with means and percentages; a single item is ordered-category data. Its mean needs the response scale and should be read alongside the response distribution.

Nami (2020). Educational smartphone apps for language learning in higher education: Students' choices and perceptions
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Pause and explain

Does 86.3% agreement show that the apps improved vocabulary scores?

Compare your explanation

No. It describes students' perceptions of usefulness. This table does not report a tested vocabulary gain.

Textbook reading: Fein et al. (2022), §§1.1-1.2 (pp. 3-5; PDF pp. 11-13); Larson-Hall (2009), Chapters 2-3. Open the free Fein et al. textbook

Read three things in a test result

How big is the pattern? How uncertain is it? What does the test ask?

The null hypothesis is the reference claim tested: for example, no mean difference or no correlation in the population. Each method defines the claim precisely.

Read	Plain meaning
Estimate / effect size	The direction and size of a difference or association. A difference of 2 points and a correlation of .50 are different quantities.
Confidence interval (CI)	An interval expressing uncertainty about an estimate under the analysis assumptions. [0.5, 3.5] gives lower and upper limits; it is not the range of individual scores.
p-value	Assuming the stated null hypothesis and model, how unusual would a test statistic at least this extreme be? Compare with the planned threshold, often .05.

$p < .05$ does not tell us that the effect is large, useful or causal. $p \geq .05$ does not prove there is no effect. The design, estimate and uncertainty still matter.

A sample is the set actually studied. A population is the wider set the question concerns. Statistical testing does not repair a biased sample or a poor measure.

Pause and explain

If one result has a smaller p-value, must its effect be larger?

Compare your explanation

No. Sample size and variability also affect p. Read the actual estimate or named effect size.

Textbook reading: Larson-Hall (2009), §§4.1, 4.3-4.4 (pp. 97-104, 114-124); Fein et al. (2022), Chapter 2. Open the free Fein et al. textbook

t-tests: which two means?

A t-test compares means on the same score scale. First decide whether the observations come from different people or the same people.

Design	Question	What goes into the test?
Independent-samples t-test	Do two separate groups have different mean scores?	One score per person; each person belongs to one group.
Paired-samples t-test	Does the same group change on average?	Two linked scores per person; analyse each person's change.

For example, comparing separate quiet-room and music-room learners is independent. Measuring everyone before and after a course is paired. Two columns of scores alone do not tell you which design you have.

Pause and explain

Twenty learners take the same test before and after a workshop. Which t-test fits that pairing?

Compare your explanation

A paired t-test. There are 20 linked changes, not 40 independent learners.

Textbook reading: Larson-Hall (2009), §§5.6, 9.1, 9.4-9.6 (pp. 136-138, 241-243, 254-264); Fein et al. (2022), §§3.1-3.4 (pp. 20-26).
Open the free Fein et al. textbook

A difference needs a reference for uncertainty

Two sample means rarely match exactly. The test asks whether the observed difference is large relative to its standard error.

For the usual two-sided t-test, the null hypothesis is a population mean difference of zero. The alternative is a nonzero difference in either direction.

Think in this order

- Name the contrast: group A minus group B, or after minus before. That choice determines the sign.

- Look at the difference in the original units. A difference of 2 quiz points is easier to judge when you know the test is out of 10 or 100.

- Read the uncertainty. SD describes variation among people; SE describes sampling uncertainty of an estimate. The t statistic compares the estimated contrast with its SE.

Check the design before trusting the result

For an independent test, check whether the groups really contain independent observations and inspect group distributions and unusual scores. Welch’s t-test allows unequal group variances; the pooled version assumes equal population variances.

For a paired test, check the distribution of the differences. Its normality assumption concerns those differences, especially with small samples.

Pause and explain

Does a negative t necessarily mean the treatment was harmful?

Compare your explanation

No. It depends on the order of subtraction and what higher scores mean.

Textbook reading: Larson-Hall (2009), §§5.6, 9.1, 9.4-9.6 (pp. 136-138, 241-243, 254-264); Fein et al. (2022), §§3.1-3.4 (pp. 20-26). Open the free Fein et al. textbook

Decode a reported t-test

Read the contrast and its units before reading the p-value.

Nami compares perception scores for students using language-learning apps as well as dictionaries (group 1) and those using dictionaries only (group 2). This is a comparison of app users, not a randomised test of learning gains.

Table 4: equal-variances-assumed row, selected values

Reported output	How to read it
Mean difference = 1.864	Group 1 minus group 2 on the perception-score scale.
SE of difference = .490	Sampling uncertainty of the mean-difference estimate.
95% CI [.902, 2.827]	Both limits are positive; this interval excludes a zero mean difference.
t = 3.808; df = 379	The test statistic and its reference degrees of freedom. df is not the effect size.
Sig. (2-tailed) = .000	Rounded software output. Report p < .001; a p-value is not literally zero.

Cohen's d expresses a mean difference in standard-deviation units: d = .50 means half of the specified SD. The version used must be named, especially for paired data. Table 4 does not supply d; do not invent it. The table also reports a Welch row with a fractional df. Name the version used rather than combining numbers from different rows. A fuller report also gives each group's n, M and SD and a named effect size.

Run it in SPSS: clicks, settings and output

Follow the menu and dialog-box labels below, then connect your output with the reporting example on this page. Menu labels are kept in English so you can find them in SPSS.

Version note: Compare Means and Proportions may be called Compare Means. Effect-size, confidence-interval and homogeneity controls vary across versions; use the dialog's Help if an option is absent.

Download SPSS practice data (CSV)

36 fictional learners for practising the procedure. These are not the research papers' raw data; the results will not reproduce the papers' findings.

Prepare the data: import, coding and settings

1. **File → Import Data → CSV / CSV Data…**
Choose the downloaded file. Select First line contains variable names and the Comma delimiter. Check that the preview has seven columns and 36 data rows, then click OK.

2. **Variable View → Type / Values / Measure**
All fields should be Numeric. Use the … button in Values to add category codes and labels. Set Measure as below; this label does not establish that a statistical test is appropriate.

3. **Data → Select Cases → All cases; Split File → Analyze all cases; Weight Cases → Do not weight cases**
For this practice file, use all individual records without splitting or weighting. Restore these settings if an earlier analysis changed them; confirm each dialog with OK.

Variable name	Measure	Role and coding
id	Nominal	Learner identifier; one row per person.

Variable name	Measure	Role and coding
study_group	Nominal	Values: 1 = quiet room; 2 = music room; 3 = café.
quiz	Scale	Quiz score out of 100; use in the independent t-test, correlation and ANOVA.
before, after	Scale	The same learner's workshop scores before and after, out of 100.
study_hours	Scale	Hours studied per week; correlate with quiz.
preference	Nominal	Values: 1 = video; 2 = text. Use with study_group in Crosstabs.

IBM: Core System User's Guide - Reading CSV Files, Variable View, Select Cases, Split File and Weight Cases

Independent-samples t-test

Do quiet-room and music-room learners have different mean quiz scores?

One row per learner. quiz is the score; study_group identifies unrelated groups: 1 = quiet room, 2 = music room. The practice file also contains group 3, which this test excludes.

Menu path
Analyze → Compare Means and Proportions → Independent-Samples T Test…

Click by click

1. **Analyze → Compare Means and Proportions → Independent-Samples T Test…**
Open the independent-samples dialog. If your menu says Compare Means, use that submenu instead.

2. **Test Variable(s): quiz; Grouping Variable: study_group**
Select a variable in the left list and use the arrow to move it into its box. The score and group code go in different boxes.

3. **Define Groups… → Use specified values → Group 1: 1; Group 2: 2 → Continue**
Type the stored codes, not the labels "quiet" and "music". The comparison is group 1 minus group 2; only these 24 learners enter this test.

4. **Options… → Confidence Interval Percentage: 95 → Continue**
Keep 95% for the mean-difference CI. For one outcome, cases missing its score or group are excluded.

5. **Estimate effect sizes; Homogeneity of variance test (if shown)**
Request effect sizes when available. In version 31+, select the homogeneity checkbox for Levene's test; older versions include it automatically.

6. **OK**
The Output Viewer opens with descriptive and test tables. Keep all numbers for the chosen t-test version on the same row.

What is produced, and where do I read it?

Table / chart in Output Viewer	What to find and read
Group Statistics	Find N, Mean and Std. Deviation for each group. This practice comparison has N = 12 per group.
Independent Samples Test	Equal variances not assumed is the Welch row. Read t, df, Two-Sided p / Sig. (2-tailed), Mean Difference and both CI limits from that row when reporting Welch.
Homogeneity of Variance Test / Levene's Test	This tests equal variances, not equal means. Its Sig. is not the p-value for the mean difference.
Independent Samples Effect Sizes (when requested)	Read the named effect-size estimate and its interval. Record the standardiser and comparison direction.

Inspect group distributions and unusual scores. Welch can be specified without using a preliminary Levene p-value as an automatic switch; the pooled row requires the equal-variance assumption.

IBM procedure documentation: IBM: Independent-Samples T Test · IBM: Define Groups · IBM: T-TEST output · IBM: version 31 interface changes

Paired-samples t-test

Do the same learners' mean scores change after a workshop?

One row per learner, with before and after in two columns. Keep each person's scores on the same row; this example uses all 36 learners.

Menu path
Analyze → Compare Means and Proportions → Paired-Samples T Test…

Click by click

1. **Analyze → Compare Means and Proportions → Paired-Samples T Test…**
Open the paired dialog; use Compare Means if that is your submenu's name.

2. **Paired Variables → Variable1: after; Variable2: before**
Select the two variables and move them into one pair. Putting after first makes the reported difference after minus before.

3. **Options… → Confidence Interval Percentage: 95 → Continue**
Keep a 95% CI. A learner needs both scores to enter this paired comparison.

4. **Estimate effect sizes → Standard deviation of the difference (if available)**
For this walkthrough, use the SD of the within-person differences as the standardiser. Name this paired-d definition when reporting; it differs from independent-groups d.

5. **OK**
SPSS produces paired descriptives, a correlations table and the paired-difference test.

What is produced, and where do I read it?

Table / chart in Output Viewer	What to find and read
Paired Samples Statistics	Read Mean, N and Std. Deviation for after and before; N = 36 in the complete practice file.
Paired Samples Test	Read Paired Differences: Mean, its 95% CI, t, df and the two-sided p. With 36 complete pairs, df = 35.
Paired Samples Correlations	This shows how before and after scores correlate. Its significance does not test whether the mean changed.
Paired Samples Effect Sizes (when requested)	Read Cohen's d or Hedges' correction and its CI, checking that the selected standardiser matches your report.

Check the distribution of after − before, especially for a small sample. Neither the correlation table nor two separate SDs replace this check.

IBM procedure documentation: IBM: Paired-Samples T Test and effect-size standardisers · IBM: paired-test options · IBM: T-TEST output

Keep a reproducible record

To save the dialog settings, click Paste after configuring the main dialog. In the Syntax Editor, select the pasted commands and choose Run → Selection, then save the .sps file. OK runs immediately instead. In the Output Viewer, use File → Save As for .spv output, or File → Export for selected tables and charts.

IBM: Paste and run syntax · IBM: export output

Nami (2020). Educational smartphone apps for language learning in higher education: Students' choices and perceptions
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Textbook reading: Larson-Hall (2009), §§5.6, 9.1, 9.4-9.6 (pp. 136-138, 241-243, 254-264); Fein et al. (2022), §§3.1-3.4 (pp. 20-26).
Open the free Fein et al. textbook

Read the paper: find the two groups and the result

Use the article's Method to identify the comparison, then find the numerical result in Table 4.

Find these in the full PDF

- Who belongs to group 1 and group 2? Is the outcome a perception score or an achievement test?

- Locate the pooled and Welch rows. Underline t, df, p, the difference and the interval from the same row.

- Explain the positive contrast in one sentence, including its units. What causal claim would go beyond this comparison?

> The mean difference (1.864 (95% confidence interval (CI): 0.902 to 2.827)) is 3.80 times larger than the standard error of difference (0.490)

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Pause and explain

What can you say without claiming that apps caused better learning?

Compare your explanation

The app-plus-dictionary users had a higher mean perception score than dictionary-only users. Table 4 reports a mean difference of 1.864, 95% CI [.902, 2.827], t(379) = 3.808, p < .001 for the pooled test. This is an association between user groups and perceptions.

Textbook reading: Larson-Hall (2009), §§5.6, 9.1, 9.4-9.6 (pp. 136-138, 241-243, 254-264); Fein et al. (2022), §§3.1-3.4 (pp. 20-26).
Open the free Fein et al. textbook

Correlation: do two scores vary together?

Keep each person's two scores paired. The question concerns a relationship, not a difference between group means.

- Measure two variables on the same cases: for example, study hours and quiz scores for each learner.

- A positive association means higher values of one tend to accompany higher values of the other. A negative association means higher values tend to accompany lower values.

- Plot the paired scores. Each dot represents one case. A scatterplot can reveal a curve or an unusual case that one coefficient hides.

Pearson's r describes a straight-line pattern in continuous scores. Spearman's r_s uses ranks and describes a consistently increasing or decreasing pattern, including a curve. We focus on reading Pearson's r here.

Pause and explain

If study hours are related to marks, does r tell us how many marks one more hour produces?

Compare your explanation

No. r is a unit-free association measure, not a slope in marks per hour.

Textbook reading: Larson-Hall (2009), §§5.2, 6.1, 6.3-6.4 (pp. 131-132, 149-165); Fein et al. (2022), §§4.1-4.2 (pp. 29-32). Open the free Fein et al. textbook

The sign and size of r answer different questions

Read direction first, then closeness to a straight-line pattern.

Value	Interpretation
r = +1	An exact increasing straight-line pattern, if both variables vary.
r = −1	An exact decreasing straight-line pattern.
r near 0	Little linear association. A curved relationship may still exist.
r = .56 versus r = −.56	The same linear strength, with opposite directions.

The usual test asks whether the population correlation is zero. Its p-value addresses that null; it does not measure the strength of the relationship. Strength is read from r in context.

Check independent cases, a suitable score scale, linearity and unusual observations. The assumptions for a p-value or interval matter in addition to calculating r. A common cause or reversed direction can produce an association.

Pause and explain

Can r be 1.3? Can r = 0 establish that nothing connects the variables?

Compare your explanation

r lies between −1 and +1. Zero rules out neither a U-shaped pattern nor other nonlinear dependence.

Textbook reading: Larson-Hall (2009), §§5.2, 6.1, 6.3-6.4 (pp. 131-132, 149-165); Fein et al. (2022), §§4.1-4.2 (pp. 29-32). Open the free Fein et al. textbook

Read one cell of a correlation table

Name the pair of variables, then read r, n and the significance note.

Lee studied 89 Korean EFL undergraduates. The paper relates English-as-an-international-language (EIL) perceptions to TOEIC scores, speaking, vocabulary and informal digital learning. Here we read just the EIL-TOEIC pair.

Table 2: one selected correlation

Find	Reported information	Meaning
Variables	EIL perceptions and TOEIC scores	Two measurements for the same learners.
Sample	N = 89	The study sample; check missingness and the number contributing to an analysis.
Coefficient	r = .56	A positive linear association. Higher TOEIC scores tend to accompany higher EIL perception scores.
Table note	*p < .01	The star means p < .01 in this particular table; always read the note.

r = .56 does not mean a 56% increase in perceptions. This table does not give a confidence interval for r; do not invent one. An interval, when reported, adds information about precision.

Run it in SPSS: clicks, settings and output

Follow the menu and dialog-box labels below, then connect your output with the reporting example on this page. Menu labels are kept in English so you can find them in SPSS.

Version note: Compare Means and Proportions may be called Compare Means. Effect-size, confidence-interval and homogeneity controls vary across versions; use the dialog's Help if an option is absent.

Download SPSS practice data (CSV)

36 fictional learners for practising the procedure. These are not the research papers' raw data; the results will not reproduce the papers' findings.

Prepare the data: import, coding and settings

1. **File → Import Data → CSV / CSV Data…**
Choose the downloaded file. Select First line contains variable names and the Comma delimiter. Check that the preview has seven columns and 36 data rows, then click OK.

2. **Variable View → Type / Values / Measure**
All fields should be Numeric. Use the … button in Values to add category codes and labels. Set Measure as below; this label does not establish that a statistical test is appropriate.

3. **Data → Select Cases → All cases; Split File → Analyze all cases; Weight Cases → Do not weight cases**
For this practice file, use all individual records without splitting or weighting. Restore these settings if an earlier analysis changed them; confirm each dialog with OK.

Variable name	Measure	Role and coding
id	Nominal	Learner identifier; one row per person.
study_group	Nominal	Values: 1 = quiet room; 2 = music room; 3 = café.

Variable name	Measure	Role and coding
quiz	Scale	Quiz score out of 100; use in the independent t-test, correlation and ANOVA.
before, after	Scale	The same learner's workshop scores before and after, out of 100.
study_hours	Scale	Hours studied per week; correlate with quiz.
preference	Nominal	Values: 1 = video; 2 = text. Use with study_group in Crosstabs.

IBM: Core System User's Guide - Reading CSV Files, Variable View, Select Cases, Split File and Weight Cases

Pearson or Spearman correlation

Do learners who study longer tend to have higher quiz scores?

One row per learner; study_hours and quiz are two measurements for that same person. No grouping variable is needed.

Menu path
Analyze → Correlate → Bivariate…

Click by click

1. **Graphs → Chart Builder… → Scatter/Dot → Simple Scatter**
Drag the chart into the preview, study_hours onto X and quiz onto Y; click OK. Inspect the shape and unusual points before choosing the coefficient.

2. **Analyze → Correlate → Bivariate… → Variables: study_hours, quiz**
Select both variables and use the arrow to put them together in Variables.

3. **Correlation Coefficients: Pearson; Test of Significance: Two-tailed**
Use Pearson for the planned linear association. For a rank-based monotonic question, select Spearman instead and label it accordingly; do not choose by whichever p is smaller.

4. **Options… → Means and standard deviations (Pearson); Exclude cases pairwise → Continue**
For Pearson, request means/SD when offered. This example uses complete pairs. With several variables, pairwise deletion can give different N per correlation; listwise uses a common complete set.

5. **Confidence Interval… → Estimate confidence interval… → 95 → Continue (if available)**
Request the coefficient's CI and keep the selected method recorded. This dialog was added in 27.0.1; an older version may not have it.

6. **OK**
Read the off-diagonal cell linking study_hours and quiz, not the diagonal 1.000 for a variable correlated with itself.

What is produced, and where do I read it?

Table / chart in Output Viewer	What to find and read
Scatterplot	Look for a roughly straight pattern for Pearson, or a consistently rising/falling pattern for Spearman; inspect outliers.
Descriptive Statistics (when requested)	Read each variable's mean and SD. Its individual N can differ from the complete-pair N when data are missing.
Correlations / Nonparametric Correlations	Find Pearson Correlation or Spearman's rho, Sig. (2-tailed) and N. The complete practice data give N = 36.
Confidence interval output (when requested)	Record lower and upper bounds for that same coefficient. Do not treat a CI for a variable's mean as a CI for r.

A significant correlation does not establish causation. Report which coefficient you ran; the star beside it is not its effect size.

IBM procedure documentation: IBM: scatterplot in Chart Builder · IBM: Bivariate Correlations · IBM: correlation options and missing values · IBM: correlation confidence intervals · IBM: 27.0.1 confidence-interval addition

Keep a reproducible record

To save the dialog settings, click Paste after configuring the main dialog. In the Syntax Editor, select the pasted commands and choose Run → Selection, then save the .sps file. OK runs immediately instead. In the Output Viewer, use File → Save As for .spv output, or File → Export for selected tables and charts.

IBM: Paste and run syntax · IBM: export output

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Read the paper: trace r back to the measures

Use the Method to understand what the scores mean, not just whether the table contains stars.

Find these in the full PDF

- On PDF pages 4-5, identify the participants and how the EIL measure and TOEIC scores were obtained.

- On PDF page 6, find the EIL-TOEIC cell in Table 2. Explain the sign, size and star.

- What evidence would you need before saying TOEIC performance caused more positive perceptions?

> EIL perceptions had higher correlations with TOEIC scores (r = .56; p < .01)

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Pause and explain

Explain this result in one sentence without making a causal claim.

Compare your explanation

EIL perceptions were positively associated with TOEIC scores, r = .56, p < .01. The association alone cannot establish its causal direction or rule out other explanations.

Textbook reading: Larson-Hall (2009), §§5.2, 6.1, 6.3-6.4 (pp. 131-132, 149-165); Fein et al. (2022), §§4.1-4.2 (pp. 29-32). Open the free Fein et al. textbook

ANOVA: compare several group means

One grouping factor, several independent groups, one score to compare.

Suppose separate groups study in a traditional class, a flipped class or a MOOC. The grouping factor is instruction type; its three levels are those three groups. The outcome could be a motivation score.

“One-way” means one grouping factor, not one group. An ordinary independent-groups ANOVA compares the means. It does not analyse yes/no categories or replace a repeated-measures model for the same people.

Role	Example
Factor	Instruction type
Levels	Traditional / flipped / MOOC
Outcome	Motivation score on the same instrument
Null hypothesis	All population group means are equal

Pause and explain

Why not simply run a separate unadjusted t-test for every possible pair?

Compare your explanation

A false positive (Type I error) means finding a statistically significant difference when the population means being compared are actually equal. For example, two teaching groups may have equal population means, yet their samples happen to differ enough for a test to give $p < .05$. Testing many pairs creates more chances for at least one such error. Plan the comparisons and use an appropriate multiple-comparison adjustment.

Textbook reading: Larson-Hall (2009), §§5.7, 10.1, 10.3-10.5 (pp. 139-141, 268-287); Fein et al. (2022), §§6.1-6.2 (pp. 49-53). Open the free Fein et al. textbook

Compare differences between groups with variation within groups

Different-looking means are more convincing when people within each group are relatively consistent.

- Between-group variation concerns how far group means are from the overall mean. Within-group variation concerns how much individuals vary around their group mean.

- F compares these sources of variation using their degrees of freedom. A larger F can give stronger evidence against equal means, but its interpretation also depends on both df values.

- An overall significant F says at least some means differ. It does not tell you that every pair differs, or name the pair responsible.

Before using the result

Check independent observations, distributions and unusual scores within groups, and the equal-variance assumption for the conventional test. Welch ANOVA is an alternative when group variances differ. If the same people supply every condition, use the repeated-measures branch of the map.

Follow-up contrasts answer “which difference?”. State the comparison procedure, such as Tukey or Scheffé, and any multiplicity adjustment.

Pause and explain

An ANOVA is significant. Can you immediately say A differs from B, B from C and A from C?

Compare your explanation

No. Inspect the planned contrasts or adjusted pairwise comparisons.

Textbook reading: Larson-Hall (2009), §§5.7, 10.1, 10.3-10.5 (pp. 139-141, 268-287); Fein et al. (2022), §§6.1-6.2 (pp. 49-53). Open the free Fein et al. textbook

Read F first, then the group comparisons

A real study can show a strong overall difference while two groups remain very similar.

Table 6: motivation post-test, selected values

Group	n	M	SD
Control	40	117.32	37.43
Flipped	40	297.87	30.98
MOOC	40	295.88	30.03

The source reports F = 395.20 with between-group df = 2 and within-group df = 117. Its displayed p = 0.00 is rounded; report p < .001 rather than p = 0.

Table 7 reports Scheffé comparisons: flipped versus MOOC has a reported difference of 2.00 and p = .96. Both instruction groups differ from the control group. Therefore, the overall result does not establish a difference between every pair.

A named ANOVA effect size such as η^2 describes magnitude as a variance proportion. Table 6 does not supply one; do not treat F as an effect size. Read the questionnaire and study design before judging the size or educational meaning of these scores. Statistical significance alone does not say how transferable an intervention is.

Run it in SPSS: clicks, settings and output

Follow the menu and dialog-box labels below, then connect your output with the reporting example on this page. Menu labels are kept in English so you can find them in SPSS.

Version note: Compare Means and Proportions may be called Compare Means. Effect-size, confidence-interval and homogeneity controls vary across versions; use the dialog's Help if an option is absent.

Download SPSS practice data (CSV)

36 fictional learners for practising the procedure. These are not the research papers' raw data; the results will not reproduce the papers' findings.

Prepare the data: import, coding and settings

1. **File → Import Data → CSV / CSV Data…**
Choose the downloaded file. Select First line contains variable names and the Comma delimiter. Check that the preview has seven columns and 36 data rows, then click OK.

2. **Variable View → Type / Values / Measure**
All fields should be Numeric. Use the … button in Values to add category codes and labels. Set Measure as below; this label does not establish that a statistical test is appropriate.

3. **Data → Select Cases → All cases; Split File → Analyze all cases; Weight Cases → Do not weight cases**
For this practice file, use all individual records without splitting or weighting. Restore these settings if an earlier analysis changed them; confirm each dialog with OK.

Variable name	Measure	Role and coding
id	Nominal	Learner identifier; one row per person.
study_group	Nominal	Values: 1 = quiet room; 2 = music room; 3 = café.
quiz	Scale	Quiz score out of 100; use in the independent t-test, correlation and ANOVA.
before, after	Scale	The same learner's workshop scores before and after, out of 100.
study_hours	Scale	Hours studied per week; correlate with quiz.
preference	Nominal	Values: 1 = video; 2 = text. Use with study_group in Crosstabs.

IBM: Core System User's Guide - Reading CSV Files, Variable View, Select Cases, Split File and Weight Cases

One-way ANOVA and follow-up comparisons

Do mean quiz scores differ across quiet-room, music-room and café learners?

One row per learner. quiz is the score; study_group is one factor with three codes: 1 = quiet, 2 = music, 3 = café.

Menu path
Analyze → Compare Means and Proportions → One-Way ANOVA…

Click by click

1. **Analyze → Compare Means and Proportions → One-Way ANOVA…**
Use Compare Means if that is the submenu name in your version.

2. **Dependent List: quiz; Factor: study_group**
Move the score into Dependent List and the three-level grouping variable into Factor. Do not enter the three groups as separate score columns.

3. **Options… → Descriptive; Homogeneity of variance test; Welch; Means plot → Continue**
Request group summaries, Levene's test, a robust overall test and a means chart. Keep 95% confidence where offered.

4. **Post Hoc… → Tukey OR Games-Howell → Continue**
Choose Tukey for a justified equal-variance analysis, or Games-Howell for unequal variances alongside Welch. Use the planned procedure, not both to hunt for significance.

5. **Estimate effect size for overall tests (if available)**
Select this main-dialog option to request ANOVA Effect Sizes. Availability and wording vary across versions.

6. **OK**
SPSS produces the overall tests and the requested pairwise comparisons. Match the output table to the analysis you intend to report.

What is produced, and where do I read it?

Table / chart in Output Viewer	What to find and read
Descriptives / Means Plot	Find each group's N, mean and SD; this practice file has three groups of 12. The plot helps show direction.
Test of Homogeneity of Variances	Read the Levene result as evidence about equal variances, not as the test of mean differences. Inspect distributions and group sizes too.
ANOVA OR Robust Tests of Equality of Means	For the conventional test, read F, between/within df and Sig. from ANOVA. For Welch, read Statistic, df1, df2 and Sig. from the Welch row in Robust Tests.
Multiple Comparisons	Read the chosen method's (I) group, (J) group, Mean Difference (I−J), adjusted Sig. and CI. This identifies specific comparisons beyond the overall test.
ANOVA Effect Sizes (when requested)	Name the reported effect size and CI. The conventional ANOVA effect-size table is not automatically a Welch-specific effect-size estimate.

A significant overall test does not mean every pair differs. Keep the chosen overall test and follow-up method clear; do not copy an effect size without checking its definition.

IBM procedure documentation: IBM: One-Way ANOVA and effect-size option · IBM: ANOVA options, Levene and Welch · IBM: post hoc tests, Tukey and Games-Howell

Keep a reproducible record

To save the dialog settings, click Paste after configuring the main dialog. In the Syntax Editor, select the pasted commands and choose Run → Selection, then save the .sps file. OK runs immediately instead. In the Output Viewer, use File → Save As for .spv output, or File → Export for selected tables and charts.

IBM: Paste and run syntax · IBM: export output

Pan et al. (2022). Massive Open Online Course Versus Flipped Instruction: Impacts on Foreign Language Speaking Anxiety, Foreign Language Learning Motivation, and Learning Attitude
Read PDF pages 6, 7 · CC BY 4.0

Open paper at PDF page 7

Download full paper PDF

Textbook reading: Larson-Hall (2009), §§5.7, 10.1, 10.3-10.5 (pp. 139-141, 268-287); Fein et al. (2022), §§6.1-6.2 (pp. 49-53). Open the free Fein et al. textbook

Read the paper: connect the overall and pairwise results

Use Tables 6 and 7 together. Find the question, the three groups and the result each table answers.

Find these in the full PDF

- On PDF page 3, locate the participants and motivation instrument. What does the score measure?

- On PDF page 7, locate Table 6 and mark both df values, F and p. On the same page, find flipped versus MOOC in Table 7.

- Explain why “the three groups differ overall” and “flipped and MOOC did not differ detectably” can both be reported.

> the difference between the motivation post-tests of the three groups is noticeable as Sig. (0.00) is smaller than (0.05)

Pan et al. (2022). Massive Open Online Course Versus Flipped Instruction: Impacts on Foreign Language Speaking Anxiety, Foreign Language Learning Motivation, and Learning Attitude
Read PDF pages 3, 6, 7 · CC BY 4.0

Open paper at PDF page 7

Download full paper PDF

Pause and explain

How would you avoid overstating the ANOVA result?

Compare your explanation

The study reports an overall group difference in motivation scores, $F(2, 117) = 395.20$, $p < .001$. The Scheffé comparison did not detect a difference between flipped and MOOC groups ($p = .96$); that does not prove the two methods are equivalent. Report group summaries and the named effect size when available.

Textbook reading: Larson-Hall (2009), §§5.7, 10.1, 10.3-10.5 (pp. 139-141, 268-287); Fein et al. (2022), §§6.1-6.2 (pp. 49-53). Open the free Fein et al. textbook

Chi-square: are two categories associated?

Sometimes the outcome is a category, not a score to average. Count people in the combinations.

Ask whether two categorical variables are independent. For example: is employment status associated with a learner's participation pattern in an online course?

Put one category variable in the rows and one in the columns. Each learner contributes to one cell. This cross-tabulation is also called a contingency table.

Variables	Values	Question
Employment status	Employed / unemployed	Two categories
Participation pattern	Seldom engaging / gradually disengaging / cramming	Three categories
Analysis	Chi-square test of independence	Do the proportions across participation patterns differ by employment status?

Pause and explain

Would a t-test of the category codes answer this question?

Compare your explanation

No. The codes are labels. Use counts in a contingency table, not means of the codes.

Textbook reading: Larson-Hall (2009), §§5.5, 8.1-8.4 (pp. 135-136, 206-227); Cohen, Manion & Morrison (2018), §41.4 (pp. 789-794; PDF pp. 818-823). Open the free Fein et al. textbook

Observed counts versus expected counts

“Expected” means what the no-association model predicts from the margins, not what the teacher hopes to find.

- Observed counts are the people actually recorded in each cell. Expected counts are calculated under independence using the row and column totals.

- The chi-square statistic summarises the gaps across all cells, accounting for their expected counts. It is non-negative.

- The test asks about the table overall. Row or column percentages explain the pattern; a p-value alone does not show which categories differ.

What needs checking?

Use counts and independent cases. Inspect expected counts, especially in small tables: the approximation may be unreliable when they are too small. Use an appropriate exact or simulation-based procedure if needed.

Do not enter the same person repeatedly as if each response were a new independent case. A before/after yes-no design needs a paired categorical method.

Pause and explain

Can an expected count be 7.5 if people cannot be cut in half?

Compare your explanation

It is a model expectation, not an observed headcount. Observed counts remain whole numbers.

Textbook reading: Larson-Hall (2009), §§5.5, 8.1-8.4 (pp. 135-136, 206-227); Cohen, Manion & Morrison (2018), §41.4 (pp. 789-794; PDF pp. 818-823). Open the free Fein et al. textbook

Read the counts before reading χ^2

Keep the denominator visible. The total for this analysis may be smaller than the paper's full sample.

Table 4: observed counts, selected columns

Participation cluster	Unemployed	Employed	Total
A: seldom engaging	21	95	116
B: gradually disengaging	13	64	77
C: cramming	2	49	51
Total	36	208	244

The reported result is $\chi^2 = 6.070$, df = 2, $p < .05$; Cramér's V = .158. χ^2 and p address evidence against independence. V describes association size; it has no positive/negative direction like r.

Employment status was available for 244 of the 640 learners. The test uses 244, not 640. In cluster C, the source reports expected counts of 7.5 unemployed and 43.5 employed learners, compared with observed counts of 2 and 49.

Run it in SPSS: clicks, settings and output

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36 fictional learners for practising the procedure. These are not the research papers' raw data; the results will not reproduce the papers' findings.

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3. **Data → Select Cases → All cases; Split File → Analyze all cases; Weight Cases → Do not weight cases**
For this practice file, use all individual records without splitting or weighting. Restore these settings if an earlier analysis changed them; confirm each dialog with OK.

Variable name	Measure	Role and coding
id	Nominal	Learner identifier; one row per person.
study_group	Nominal	Values: 1 = quiet room; 2 = music room; 3 = café.
quiz	Scale	Quiz score out of 100; use in the independent t-test, correlation and ANOVA.

Variable name	Measure	Role and coding
before, after	Scale	The same learner's workshop scores before and after, out of 100.
study_hours	Scale	Hours studied per week; correlate with quiz.
preference	Nominal	Values: 1 = video; 2 = text. Use with study_group in Crosstabs.

IBM: Core System User's Guide - Reading CSV Files, Variable View, Select Cases, Split File and Weight Cases

Chi-square test of independence through Crosstabs

Is study environment associated with preferring video or text?

One row per learner. study_group has three categories; preference is 1 = video, 2 = text. Use individual records without weighting.

Menu path
Analyze → Descriptive Statistics → Crosstabs…

Click by click

1. **Analyze → Descriptive Statistics → Crosstabs…**
Use this two-variable independence route. It differs from a one-variable goodness-of-fit chi-square procedure.

2. **Row(s): study_group; Column(s): preference**
Move the group code into rows and preference into columns. With this layout, row percentages compare preferences within each group.

3. **Statistics… → Chi-square; Phi and Cramer's V → Continue**
Request the test and an association-size measure. These produce Chi-Square Tests and Symmetric Measures.

4. **Cells… → Counts: Observed, Expected; Percentages: Row → Continue**
Show actual counts, the no-association baseline and each group's percentages. Leave Create APA style table off if it disables those cell choices.

5. **OK**
The practice crosstab is 3 × 2. Each group has 12 people; there are 18 video and 18 text preferences, so every expected count is 6.

6. **Chi-Square Tests → expected-count footnote; Exact… if needed and available**
Inspect expected counts before using the asymptotic p-value. Sparse tables may need Fisher's exact test (2 × 2) or an exact/Monte Carlo procedure; availability depends on edition and platform.

What is produced, and where do I read it?

Table / chart in Output Viewer	What to find and read
Case Processing Summary	Check Valid N and Missing. This complete practice dataset should give Valid N = 36.
study_group × preference Crosstabulation	Compare Count, Expected Count and % within study_group. Row percentages use that row's 12 learners as denominator.
Chi-Square Tests → Pearson Chi-Square	Read Value (χ^2), df and Asymptotic Significance (2-sided). Here df = (3−1)(2−1) = 2. Read the expected-count footnote below the table.
Symmetric Measures → Cramer's V	Read the Value as association size. Use the crosstab percentages for the pattern; V has no positive/negative direction.

For this practice, observed video/text counts are 9/3, 6/6 and 3/9, but expected counts are 6/6 in every group. Expected means the no-association baseline, not a target someone hopes to reach.

Do not enter before/after responses from the same person as unrelated cases. A paired yes/no question needs an appropriate paired categorical analysis.

IBM procedure documentation: IBM: running Crosstabs · IBM: Crosstabs statistics and Cramér's V · IBM: cell counts and percentages · IBM: obtaining exact tests

Keep a reproducible record

To save the dialog settings, click Paste after configuring the main dialog. In the Syntax Editor, select the pasted commands and choose Run → Selection, then save the .sps file. OK runs immediately instead. In the Output Viewer, use File → Save As for .spv output, or File → Export for selected tables and charts.

IBM: Paste and run syntax · IBM: export output

Tang & Xing (2022). Massive open online courses for professional certificate programs? Perspectives on professional learners' longitudinal participation patterns
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Textbook reading: Larson-Hall (2009), §§5.5, 8.1-8.4 (pp. 135-136, 206-227); Cohen, Manion & Morrison (2018), §41.4 (pp. 789-794; PDF pp. 818-823). Open the free Fein et al. textbook

Read the paper: explain the association with proportions

Find the table, its sample and the report. Then use the appropriate denominator to explain the pattern.

Find these in the full PDF

- On PDF page 4, find how employment status was obtained and why only 244 learners have this variable.

- On PDF page 8, locate Table 4, the expected-count rows and the paragraph reporting χ^2 and Cramér's V.

- Compare the proportion in cluster C within each employment group: 49 of 208 employed learners and 2 of 36 unemployed learners. What conclusion goes beyond the data?

> The result ($\chi 2 = 6.070$; $df = 2$; $p < .05$)

Tang & Xing (2022). Massive open online courses for professional certificate programs? Perspectives on professional learners' longitudinal participation patterns
Read PDF pages 4, 8 · CC BY-NC-ND 4.0

Open paper at PDF page 8

Download full paper PDF

Pause and explain

Why is "49 employed versus 2 unemployed" not enough by itself?

Compare your explanation

The groups have different totals. About 23.6% of employed learners and 5.6% of unemployed learners fall in C (percentages calculated from Table 4). The paper reports an association, $\chi^2(2, N = 244) = 6.070$, $p < .05$, $V = .158$; this does not establish that employment caused cramming.

Textbook reading: Larson-Hall (2009), §§5.5, 8.1-8.4 (pp. 135-136, 206-227); Cohen, Manion & Morrison (2018), §41.4 (pp. 789-794; PDF pp. 818-823). Open the free Fein et al. textbook

Quantitative data analysis: method map

We have studied four methods together: t-tests, correlation, one-way ANOVA and chi-square. Use this map to explore other methods at your own pace: find your question and data type, open a method, then return here to compare another. The next lesson page is the glossary.

Main categorisation reference: UCLA OARC · Choosing the Correct Statistical Test. Advanced branches extend the map by question and data structure; each choice still requires checks of the model assumptions.

Choose by outcomes and predictors

Describe one sample

What does the sample distribution show?

Outcomes

One or more · Continuous or categorical

Predictors

None required

- Descriptive statistics

One outcome: compare groups

Do group means or rank distributions differ?

Outcomes

One · Continuous or ordinal

Predictors

Categorical groups; optional covariates

- Independent and paired t-tests
- One-way and factorial ANOVA
- ANCOVA
- Mann-Whitney, Wilcoxon and Kruskal-Wallis tests
- Equivalence testing / TOST

One categorical outcome

Are categories associated, or what predicts category probabilities?

Outcomes

One · Binary, ordered or unordered categories

Predictors

Categorical, continuous or mixed

- Chi-square tests and contingency tables
- Binary logistic regression
- Ordinal and multinomial regression

One continuous outcome: association or regression

How does a continuous outcome vary with the predictors?

Outcomes

One · Continuous

Predictors

Continuous, categorical or mixed

- Pearson, Spearman and partial correlation
- Linear, multiple and hierarchical regression
- Generalised additive models (GAMs)
- Robust regression

Several continuous outcomes

Do groups differ across outcomes considered jointly?

Outcomes

Two or more · Several continuous outcomes

Predictors

Categorical groups; optional covariates

- MANOVA and MANCOVA

Many indicators: dimensions and measurement

What structure underlies the indicators?

Outcomes

No single designated outcome · Continuous or categorical indicators

Predictors

Several observed indicators

- Exploratory factor analysis
- Confirmatory factor analysis
- Principal component analysis
- Measurement invariance / multi-group CFA
- Internal consistency and rater agreement
- Item response theory: 2PL and polytomous models

Repeated outcomes across time

How do outcomes change or relate across occasions?

Outcomes

One or more, repeatedly measured · Repeated continuous outcomes or time series

Predictors

Time, groups or prior measurements

- Repeated-measures and mixed ANOVA
- Latent growth curve modelling
- Cross-lagged panel models
- Random-intercept CLPM
- Time-series analysis

Extend for the question and data structure

Indirect effects, interactions and SEM

Through which pathway, or under which conditions, are variables related?

Outcomes

One or a system of outcomes · Observed or latent outcomes

Predictors

Predictors, mediators, moderators or latent constructs

- Mediation and indirect effects
- Moderation and interaction models
- Moderated mediation / conditional process
- Structural equation and path modelling

Event counts and rates

What predicts the expected count or rate?

Outcomes

One · Non-negative counts; exposure if needed

Predictors

Continuous, categorical or mixed

- Count regression (Poisson / negative binomial)

Clusters, latent profiles and latent classes

Which cases show similar multivariate patterns?

Outcomes

No designated outcome for grouping · Continuous for LPA; categorical for LCA

Predictors

Several grouping indicators

- K-means and hierarchical clustering
- Latent profile analysis
- Latent class analysis (LCA)

Nested, crossed or otherwise correlated observations

How should the model represent dependence?

Outcomes

One or more correlated outcomes · Match the response distribution and link

Predictors

Predictors plus person/group/item structure

- Linear / generalised mixed-effects models

- Generalised estimating equations

Time until an event

How long until an event occurs?

Outcomes

One event-time outcome · Time plus event/censoring status

Predictors

Continuous, categorical or mixed

- Survival / time-to-event analysis

Conditional relationships in a system

Which variables remain associated conditional on the others?

Outcomes

No single designated outcome · Joint multivariate observations

Predictors

Variables treated as nodes

- Network analysis: conditional-association models

Prediction for unseen cases

How accurate are predictions on held-out data?

Outcomes

One or more · Continuous or categorical

Predictors

Features with independent validation

- Random forests and boosted trees

- Support vector machines

- Neural networks for prediction

Effect estimates from several studies

What do study estimates show together?

Outcomes

Study effect estimates · Effect estimates and sampling variances

Predictors

Study characteristics or none

- Random-effects meta-analysis
- Meta-regression and moderator analysis
- Three-level meta-analysis / robust variance estimation

Estimation and uncertainty across models

How will parameters and uncertainty be estimated?

Outcomes

Defined by the substantive model · Must respect the model and sampling structure

Predictors

Defined by the substantive model

- Bayesian estimation
- Bootstrap / resampling inference

Statistical glossary

106 terms: start with describing and comparing data, continue with common reporting notation, then consult terms for particular analyses. Definitions are written out within each category.

Click − to collapse a category or + to expand it. Several categories can stay open together.

The third column distinguishes mathematical bounds, no-effect references and conventional benchmarks. Some terms have no numerical range or universal cutoff; interpretation depends on the model, scale, design and research context.

The fourth column explains every term through familiar situations such as commuting, cafés, studying and sleep. Numbers are illustrative.

Start with describing and comparing

Describing data

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
n: sample size	The number of observations or units included in the stated analysis. Check the unit: 300 ratings from 30 people do not represent 300 independent people. Also written as: n · N	**Positive whole number** n = 80 means 80 stated units. Identify whether these are people, responses or studies; there is no universal minimum sample size.	You ask 30 classmates how they travel to campus. Here n = 30 counts people, not the number of questions they answer.
$\bar{x}$ / M: sample mean	Add the observed values and divide by their number. For 2, 4, 4, 6, the mean is 16/4=4. In descriptive tables M often means mean; in a mediation diagram M instead labels the mediator. Also written as: mean · sample mean · $\bar{x}$ · M	**Within the measurement scale** A mean on a 1-5 scale lies between 1 and 5. It can be negative if the scale permits negative values; there is no universal “good” mean.	Three friends spend $20, $30 and $40 on lunch. Their mean spending is $30: the total shared equally among three people.
Median	The middle value after sorting observations, or the mean of the middle two when their number is even. It describes location and can differ from the mean in a skewed distribution. Also written as: median	**Within the measurement scale** At least 50% of the distribution is at or below the median and at least 50% at or above it. Its possible numerical values follow the score scale.	Five delivery times are 10, 12, 15, 18 and 60 minutes. The median is the middle one, 15 minutes; one very late order does not move it to 60.
SD: standard deviation	Describes the spread of observations around their mean, in the original measurement units. SD is the square root of variance. It is different from the standard error of an estimate. Also written as: standard deviation · standard deviations · SD	**SD ≥ 0** 0 means no observed spread. SD = 2 means a spread measured in two original score units; its size is meaningful relative to the scale.	Two cafés both average a 10-minute wait. In one, waits cluster near 10; in the other, they vary widely. SD describes that spread in minutes.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Variance	A measure of spread based on squared deviations. Sample variance usually divides the sum of squared deviations from the sample mean by n−1. Variance uses squared units; SD returns to the original units. Also written as: variance · variances	**Variance ≥ 0** 0 means no variation. Variance = 4 corresponds to SD = 2; variance uses squared units. A negative fitted variance component requires investigation.	For café waiting times, a variance of 4 minutes squared corresponds to an SD of 2 minutes. Both describe spread; variance uses squared units.
IQR / Q1 / Q3	The interquartile range is Q3−Q1: the spread of the middle 50% of observations. Q1 and Q3 are the 25th and 75th percentiles. IQR describes observations, rather than the uncertainty of a mean estimate. Also written as: IQR · quartile · Q1 · Q3	**IQR ≥ 0; Q1 ≤ Q3** Q1 and Q3 mark the 25th and 75th percentiles. IQR = 2 means the middle half of observations spans two scale units.	For a group of commuters, Q1 is 20 minutes and Q3 is 35. The middle half of journey times spans 15 minutes: that is the IQR.
Range / minimum / maximum	Minimum and maximum are the smallest and largest observed values. Range=maximum−minimum. They describe the observed sample; they are different from the lower and upper limits of a confidence interval. Also written as: range · minimum · maximum	**Range ≥ 0; minimum ≤ maximum** With minimum 2 and maximum 8, range = 6. Minimum and maximum can be negative; they are not confidence limits.	Your shortest journey this week took 15 minutes and your longest took 45. The range is 30 minutes: longest minus shortest.
σ^2 / σ: variance and SD parameters	σ^2 often denotes a population or model variance, and σ its standard deviation. Subscripts specify the component, such as residual variation. Check the model's units and component definitions. Also written as: σ^2 · σ	**$\sigma \geq 0$; $\sigma^2 \geq 0$** $\sigma = 2$ corresponds to $\sigma^2 = 4$. There is no shared upper bound across measurement scales.	If we describe waiting times for all customers, σ represents their population SD; σ^2 represents the corresponding variance. A sample gives estimates of these quantities.
μ: expected value	A model mean or expected value. In count regression μ is the expected count; in growth models μ with a subscript may denote a mean intercept or mean slope. Also written as: μ	**Follows the outcome/model domain** A Gaussian mean can be any real value; a count mean is non-negative, and a binary-response mean is a probability between 0 and 1.	A bus may average 2.5 late arrivals per week over many weeks. $\mu = 2.5$ is an expected average, even though any one week has a whole-number count.
Σ: sum	Add the indicated quantities over the specified index. For example, Σx adds all the x values. The index and limits identify which observations or components are included. Also written as: Σ	**No fixed range** Σ is an operation, not one statistic with a cutoff. A sum of signed values can be negative; a sum of squares cannot.	Add Monday's 20-minute walk, Tuesday's 30 and Wednesday's 10. Σ says "add them all": the total is 60 minutes.
Standardised	Expressed on a rescaled metric, often using standard deviations. A standardised variable $z=(x-\text{mean})/SD$ has mean 0 and SD 1 in the sample used for the transformation. State which variables or coefficients were standardised. Also written as: standardised · standardized	**Depends on the quantity standardised** Standardising does not generally confine values to −1 to +1. For z scores, 0 is the mean and 1 is one SD above it.	Height is measured in centimetres and test scores in points. Standardising expresses each relative to its own group's mean and SD, so the scales can be compared.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
z: standardised score	Here z denotes a standardised variable: (value−mean)/SD. A value of 1 is one SD above the mean; −.5 is half an SD below it. This is different from a coefficient's z-test statistic. Also written as: $z_1 \cdot z_2$	**Any real value: −∞ to +∞** 0 = sample mean; +1/−1 = one SD above/below. For an approximately normal distribution, about 95% fall within ±1.96; these are not absolute score limits. NIST: normal critical values	Your class averages 70 points with SD = 10. A score of 80 has z = 1: one SD above the class mean.

Comparing groups

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Mean difference	The difference between two specified means. Define its direction: mean(A)−mean(B)=3 means A averages three outcome units higher. For paired data, analyse within-pair differences and their uncertainty. Also written as: mean difference	**Can be negative, zero or positive** For A−B, +3 means A is three units higher; −3 means lower. 0 is the no-difference reference; practical importance depends on the scale and interval.	Quiet-room learners average 75 points and music-room learners average 70. Quiet minus music gives a mean difference of +5 points.
Homogeneity of variance	An assumption that the groups have equal population variances: the same spread, rather than the same mean. The pooled independent-samples t-test and conventional one-way ANOVA use this assumption. Welch procedures allow unequal group variances. Equal sample SDs are not required in every realised sample. Also written as: homogeneity of variance · homogeneity of variances · equal variances · variance homogeneity	**Equal positive population variances give a variance ratio of 1** This is a model assumption, not a statistic with a universal cutoff. Sample variances can differ even when population variances are equal. Inspect group spreads and the design; Welch methods do not require equal variances. NIST: equal variances	Two classes could have different averages but similarly spread-out scores. Equal population spread is the assumption; equal averages are a separate question.
t statistic	For a coefficient or mean difference, t=(estimate−null value)/SE. Compare it with the appropriate t distribution and degrees of freedom. In time-indexed equations, t can instead label an occasion. Also written as: t	**Any real value** 0 is the null-centred reference. Sign follows the contrast; larger |t| gives a smaller two-sided p at fixed df. Critical values depend on df and the chosen α, not one universal t cutoff.	Two classes differ by 4 points, with SE = 2 for that difference. t = 2 means the gap is two standard errors from the no-difference value of zero.
F statistic	A ratio of model-defined mean squares or a joint-test statistic. In a simple ANOVA, F=MS_between/MS_within. Read both numerator and denominator df; a significant omnibus test does not identify the differing groups. Also written as: F	**F ≥ 0** In ordinary ANOVA, F near 1 means comparable effect and error mean squares. The upper-tail cutoff depends on both df and α; F is not a percentage or an effect size.	Compare three cafés' waiting times. F weighs how far their average waits differ against how much customers' waits vary within each café.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
df: degrees of freedom	An index used in the statistic's reference distribution, reflecting information remaining after constraints or fitted parameters. For a paired t-test on n differences, df=n−1. It is not generally the sample size. Also written as: degrees of freedom · df	**Usually ≥ 0; can be non-integer** Paired t: df = n−1. Welch approximations may use fractional df. SEM df = 0 denotes a saturated/just-identified specification whose global fit cannot test restrictions.	Three friends' spending totals $90. Once two amounts are fixed, the third must make up the total. Only two amounts can vary freely: df = 2 in this example.
Effect size	A defined measure of the magnitude of a difference or association. Examples include mean differences, standardised differences and correlations. Interpret the units, uncertainty and substantive context alongside p-values. Also written as: effect size · effect sizes	**Depends on the effect-size definition** The no-effect reference may be 0 (mean difference, r, d) or 1 (OR, IRR, HR). There is no numerical cutoff shared by all effect sizes.	A queue system saves customers 3 minutes on average. Those 3 minutes describe the size of the benefit; a p-value answers a different question.
d: standardised mean difference	A mean difference divided by a specified SD. d=.50 means half of that SD, rather than a 50% improvement. Paired and independent-group versions use different standardisers, which must be stated. Also written as: Cohen d · Cohen's d · Cohen's d · standardised mean difference	**Signed; no universal ±1 bound** 0 = no standardised mean difference. Conventional |d| anchors .20/.50/.80 are called small/medium/large; they are rough context-dependent benchmarks, not significance cutoffs. Cohen: conventional effect sizes	Two classes differ by 5 points. If the chosen comparison SD is 10, d = .50: the gap is half an SD, not a 50% score increase.
g: Hedges' g	A standardised mean-difference estimator with a small-sample bias correction. State the effect-size definition, comparison direction, uncertainty and treatment of dependent data. Also written as: Hedges g · Hedges' g · Hedges' g	**Signed; no universal ±1 bound** 0 is the no-difference reference. The .20/.50/.80 standardised-difference anchors are sometimes applied, but g includes a bias correction and should be interpreted with its interval and field context. Cohen: conventional effect sizes	You compare two small tutoring groups. Hedges' g expresses their mean gap in SD units and applies a correction for small-sample bias.
χ^2: chi-square statistic	A statistic evaluated against a specified chi-square reference distribution. In a contingency table it summarises discrepancies between observed and expected counts; in SEM it assesses model discrepancy. Its df and model context matter. Also written as: χ^2 · chi-square statistic	**$\chi^2 \geq 0$ for the usual statistic** 0 means no observed discrepancy on that statistic. Larger values indicate more discrepancy; compare with the df-specific reference distribution, not a universal number such as 3.84.	Each of two classes has 10 students; 10 students overall prefer video. With no class-preference link, expect 5 in each. Observing 9 and 1 creates gaps that χ^2 summarises.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Cramér's V	A measure of association size for a contingency table, derived from chi-square, the sample size and table dimensions. It ranges from 0 to 1. It has no positive/negative direction; use proportions to describe the pattern. Its magnitude is separate from the p-value and does not establish causation. Also written as: Cramér's V · Cramér V · Cramers V · Cramer's V · V	**0 ≤ V ≤ 1** 0 indicates no association on this measure; larger values indicate stronger association. There is no sign or universal significance cutoff. Small/medium/large conventions depend on table dimensions and context.	After testing whether class and video/text preference are related, Cramér's V describes the association's size. The percentages show which class prefers what.
Contingency table / cross-tabulation	A table of counts for combinations of categorical variables. Identify what rows and columns mean, give totals and denominators, and distinguish observed counts from counts expected under the specified model. Also written as: contingency table · contingency tables · cross-tabulation · cross-tabulations	**Observed counts: non-negative whole numbers** Expected counts can be fractional. Percentages require a stated row, column or overall denominator. A 2×3 independence table has (2−1)(3−1)=2 degrees of freedom.	Make a table with Class A/B as rows and video/text as columns. Each cell counts students with that combination: a cross-tabulation.
SS: sum of squares	A sum of squared deviations allocated to a model component. Dividing SS by its df gives a mean square (MS). The subscripts identify the component, such as between or within groups. Also written as: SS	**SS ≥ 0 for a sum of squares** 0 means all contributing deviations are zero. Raw SS grows with scale and sample size, so it has no common "large" cutoff.	For café waits, find each customer's distance from a relevant mean, square it and add the results. SS collects these squared gaps into a total.
MS: mean square	A sum of squares divided by its associated degrees of freedom: MS=SS/df. ANOVA F ratios compare particular mean squares. Also written as: MS	**MS ≥ 0 when df > 0** MS = SS/df. It is in squared outcome units; compare the appropriate mean squares rather than interpreting one MS against a fixed threshold.	In the café comparison, divide each relevant SS by its df. This puts each total on a per-degree-of-freedom basis; F then compares the resulting mean squares.
η^2: eta squared	An ANOVA effect-size measure: SS_effect/SS_total. It expresses the proportion of sample total variation attributed to that effect. Partial η^2 uses a different denominator. Also written as: η^2	**$0 \le \eta^2 \le 1$ for the usual SS ratio** η^2 = .06 means 6% of the sample total variation is attributed to the specified effect. Rough .01/.06/.14 benchmarks are sometimes labelled small/medium/large; design and context matter.	For three study groups, η^2 = .10 means 10% of the sample's total score variation is attributed to the grouping factor in this ANOVA.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
ω²: omega squared	An ANOVA effect size estimating the proportion of population outcome variation associated with the specified factor. It corrects some of the upward sampling bias in η². In a one-way ANOVA, ω² = .06 estimates about 6% of outcome variation associated with group membership. The population proportion is 0-1; sample estimates can be negative near zero, and some software reports these as 0. This is not the reliability coefficient ω and does not itself establish causation. Also written as: ω² · ω2 · omega squared · omega-squared	**Population proportion: 0-1; sample estimates may be negative** ω² = .06 estimates about 6% of outcome variation associated with the factor in a one-way ANOVA. Negative sample estimates indicate a value near zero, not a negative amount of variance; state any truncation to 0. Rough .01/.06/.14 small/medium/large benchmarks are sometimes used, but conventions differ: name the guideline and interpret the study context. These are not p-value cutoffs. effectsize: ANOVA effect sizes · effectsize: interpretation conventions · effectsize: negative estimates and zero truncation	For the same study groups, ω² estimates how much population score variation is associated with grouping, correcting some upward bias in sample η².
ηp²: partial eta squared	SS_effect/(SS_effect+SS_error) for the specified effect and error term. It uses a different denominator from η² and is not generally the proportion of all outcome variance explained. Also written as: partial eta squared · partial η² · ηp²	**0 ≤ ηp² ≤ 1 for the usual SS ratio** ηp² = .06 means 6% of effect-plus-associated-error SS, not 6% of all outcome variation. Values depend on the design; avoid transferring size labels mechanically across designs.	A study includes teaching method and room type. Partial η² compares the teaching-method component with that component plus its error, rather than with all score variation.

Frequently used in Results sections

Common reporting: p, SE and confidence intervals

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
p-value	Under the null hypothesis and specified model, the probability of a test statistic at least as extreme as the observed one. p=.02 means 2% under those assumptions; it is not a 2% probability that the null is true, and does not measure effect size. The notation p < .001 means this tail probability is below 0.1%, under the null and specified model; it is not the probability that the null hypothesis is true. Also written as: p-value · p-values · p value · p	**0 ≤ p ≤ 1** p < .05 / .01 / .001 means a null-model tail probability below 5% / 1% / 0.1%. These are conventional reporting thresholds; compare with the prespecified α. Neither small p nor p ≥ .05 establishes effect size or proves a hypothesis. ASA: interpreting p-values	Suppose two teaching methods truly have equal population means. p = .02 says gaps at least this extreme would occur 2% of the time under that null model and test.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
SE: standard error	The estimated sampling variability of an estimate. SE describes precision of the estimate, whereas SD describes spread among observations. A simple large-sample 95% Wald CI is estimate ± 1.96×SE. SE = 0.50 means the estimated standard error is 0.50 in the estimate's units; for a regression coefficient those units are outcome units per predictor unit. Also written as: standard error · standard errors · SE	**SE ≥ 0; no fixed upper bound** Smaller SE means greater precision for the same parameter and units. SE is not judged against .05; that threshold concerns p. SE = 0 may indicate a degenerate or numerical situation.	Survey a different sample of commuters and its mean journey time will change. SE estimates how much that sample mean varies across repeated samples.
CI: confidence interval	A lower and upper limit expressing uncertainty about a parameter. Under its assumptions, a 95% CI procedure covers the true parameter in 95% of repeated samples. [1.02, 2.98] lists lower=1.02 and upper=2.98; it is not a range containing 95% of individuals. With an estimate of 2.00 and SE of 0.50, a large-sample normal-reference Wald calculation gives 2.00 ± 1.96 × 0.50 = [1.02, 2.98]. A t-based interval instead uses a multiplier determined by the degrees of freedom. Also written as: confidence interval · confidence intervals · CI	**Limits follow the parameter's scale** Read the confidence level (often 95%) and both limits. For a matched two-sided 5% test, a 95% CI excluding the null corresponds to rejection: null = 0 for differences/slopes, 1 for ratios. This correspondence requires the same inferential method.	A survey estimates a mean commute of 30 minutes, with 95% CI [28, 32]. The interval describes uncertainty about the population mean; individuals can travel longer or shorter.
[lower, upper]: interval notation	In 95% CI [1.02, 2.98], 1.02 is the lower limit and 2.98 is the upper limit, in the units of the parameter. These are interval endpoints, rather than two separate estimates or the range of 95% of individual scores. Read the label: square brackets can also show equivalence bounds or other intervals. Also written as: square brackets · [lower, upper]	**Lower limit ≤ upper limit** [1.02, 2.98] means lower 1.02 and upper 2.98. For a coefficient the interval excludes 0; for a ratio it includes 1 only if 1 lies between its limits. Always identify the labelled parameter and interval type.	For mean commute time, [28, 32] means a lower limit of 28 and an upper limit of 32 minutes. The brackets hold the interval's two endpoints.
LL / lower limit	The lower endpoint of the labelled interval. In CI [1.02, 2.98], LL=1.02. The lower limit is not the smallest observed individual score. Also written as: LL · lower limit	**LL ≤ UL; sign follows the parameter** A negative lower limit is possible for a signed coefficient. A ratio's admissible limits are non-negative; LL itself has no universal pass/fail value.	In a mean-commute CI [28, 32], LL = 28 minutes. It is the interval's lower endpoint, not the shortest journey anyone made.
UL / upper limit	The upper endpoint of the labelled interval. In CI [1.02, 2.98], UL=2.98. Read it together with the lower limit, estimate and stated confidence level. Also written as: UL · upper limit	**UL ≥ LL; sign follows the parameter** Read together with LL. A narrow interval indicates greater precision on the same scale; a large positive UL alone does not establish an effect.	In that mean-commute CI [28, 32], UL = 32 minutes. It is the upper endpoint, not the longest individual journey.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
z statistic	A common Wald z statistic is (estimate − null value)/SE. Its magnitude expresses distance from the null in standard-error units. A reference standard normal distribution is used when the approximation is justified. z is not an effect size. Against a null coefficient of 0, an estimate of 2.00 with SE = 0.50 gives z = (2.00−0)/0.50 = 4.00: four standard errors from zero. In a neural network z can instead mean the weighted input plus bias before activation; a standardised observation is another use of z. Also written as: z	**Any real value** For a justified standard-normal two-sided test: |z| ≈ 1.960 / 2.576 / 3.291 corresponds to p ≈ .05 / .01 / .001. These are reference-specific critical values, not effect-size labels. NIST: normal critical values	A large-sample estimate of a 4-minute queue reduction has SE = 1. Against no reduction, z = 4: the estimate is four standard errors from zero.
p = .000 in an output	Usually a display rounded to three decimal places, not a probability known to be exactly zero. Use a justified reporting convention such as p<.001 when the underlying value supports it. Also written as: p=.000 · p = .000 · rounding	**Rounded display, not a special p range** p = .000 usually means the display has rounded a small positive value. Report p < .001 when supported; do not interpret it as zero probability. ASA: interpreting p-values	Your software displays p = .000 for a café comparison because it keeps only three decimal places. Report p < .001 when supported; the probability is not exactly zero.
H_0: null hypothesis	The reference claim tested by the specified procedure, for example a mean difference of 0 or an odds ratio of 1. The null value depends on the parameter and the test. Also written as: H0 · H_0 · null hypothesis	**A claim, not a bounded statistic** Common null values: difference/slope/r = 0; ratio = 1. Define the actual parameter and null before looking at p.	When comparing two cafés, the null hypothesis could be “their population mean waiting times are equal”. This is the starting claim the test evaluates.
H_1 / H_a: alternative hypothesis	The specified alternative to H_0: for example a difference other than 0, or a difference greater than 0. A significant result does not assign a probability that this alternative is true. Also written as: H1 · H_1 · H_a · alternative hypothesis	**A claim, not a bounded statistic** A two-sided alternative may be b ≠ 0; a directional alternative b > 0. A small p does not provide P(H_1 is true).	For those cafés, a two-sided alternative is “their population mean waiting times differ”. It allows either café to have the longer wait.
α: significance level	A prespecified long-run Type I error threshold for the test, often .05. Compare p with the chosen threshold, while still interpreting magnitude, uncertainty, multiplicity and design. This α differs from reliability alpha. Also written as: significance level	**0 < α < 1 in ordinary testing** Common choices .05, .01 and .001 target long-run Type I error rates of 5%, 1% and 0.1% under the test assumptions. Choose α in advance, accounting for multiple testing where needed. ASA: interpreting p-values	Before comparing cafés, choose α = .05. It sets the long-run false-positive threshold for this test, assuming equal population means and the test’s assumptions.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Critical value / ±	A reference-distribution cutoff associated with the chosen tail probability. In estimate ±1.96×SE, ± means compute both minus and plus endpoints. The value 1.96 is a normal-reference multiplier; a t-based interval uses a df-specific multiplier. Also written as: critical value · ±	**Depends on distribution, tails, df and α** Normal reference: 95% two-sided uses ±1.96; a 5% upper one-sided test uses +1.645. A t test uses a df-specific value. ± instructs calculating both minus and plus limits. NIST: normal critical values	Think of a test's critical value as a pre-agreed boundary. For a two-sided normal z test at α = .05, values beyond about ±1.96 cross that boundary.

Correlation and regression

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
r: correlation	Pearson r expresses linear association on a scale from −1 to +1. Sign gives direction; magnitude indicates strength of that linear pattern. A correlation does not establish causality. Also written as: r	**−1 ≤ r ≤ +1** 0 = no linear association; ±1 = perfect linear association. Conventional |r| anchors .10/.30/.50 indicate small/medium/large in some settings; they are not universal field standards or p-value thresholds. Cohen: conventional effect sizes	Students who sleep longer may tend to have higher quiz scores. A positive r summarises that straight-line pattern; it does not show that sleep caused the scores.
r_s / ρ_s: Spearman correlation	Spearman correlation is Pearson correlation applied to ranks: positions after sorting the values, with average ranks for ties. It describes whether higher values of one variable consistently accompany higher or lower values of the other, including a curved pattern that keeps increasing or decreasing. This is called a monotonic relationship. A U-shaped relationship need not have a large Spearman correlation. Also written as: r_s · ρ_s · Spearman · Spearman correlation · rank · ranks · monotonic	**−1 to +1; undefined if either variable is constant** +1 means perfectly matching ranks; −1 means perfectly reversed ranks. 0 means no rank correlation, not necessarily no relationship. Test against 0 using an appropriate procedure; significance also depends on sample size and ties.	Rank friends by weekly exercise and by reported energy. Spearman asks whether those higher on one list also tend to be higher on the other.
R^2: coefficient of determination	In ordinary linear regression, the proportion of sample outcome variation accounted for relative to an intercept-only model. R^2=.40 means 40% on that definition. Other models use different pseudo-R^2 measures. Also written as: R^2 · R-squared	**Usually 0-1 for in-sample OLS with an intercept** R^2 = .40 means 40% of sample outcome variation accounted for on that definition. Predictive/out-of-sample or no-intercept R^2 can be negative. No universal "good R^2" applies across tasks. scikit-learn: predictive R^2	A model uses study hours to predict quiz scores. R^2 = .40 means it accounts for 40% of sample score variation relative to predicting everyone's mean.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Coefficient	A model parameter describing a specified relationship. Read its sign, size, units and uncertainty. A coefficient is not automatically a causal effect. Also written as: coefficient · coefficients	**Usually signed; range depends on the model** Read the defined scale. 0 often means no conditional association on the coefficient scale; size is not judged by p cutoffs or one standard numerical benchmark.	A model predicts quiz scores from study hours. The coefficient beside study hours tells you the modelled score difference associated with one additional hour.
b: unstandardised coefficient	In linear regression, b is the expected outcome-unit difference per one-unit predictor difference, holding the other included predictors constant. For example, b=2 means two outcome units per predictor unit. Its meaning changes with the model's response/link scale. Other uses in this lesson: IRT b=item difficulty; mediation b=M→Y path conditional on X; SVM b=intercept/bias; GAM b_k=basis-function coefficient. These meanings should not be interchanged. Also written as: b	**Can be negative, zero or positive** No fixed upper bound or general cutoff. b = 2 means two outcome units per predictor unit in a linear model; b = −2 reverses direction. IRT difficulty and other uses require their own scale. UCLA: regression coefficients	In a linear quiz-score model, b = 2 for study hours means one more hour is associated with 2 more points, holding the other included predictors constant.
β	In many regression outputs, β denotes a standardised coefficient: the outcome SD difference per predictor SD difference, conditional on the other predictors. In equations β may instead denote a population coefficient. Check the stated notation. In power calculations β is a Type II error probability. Beta(a,b), written as a distribution name, denotes the beta probability distribution, as in the Bayesian example. Also written as: β · beta	**Signed; not generally bounded by ±1** A standardised β = .30 means .30 outcome SD per predictor SD, conditional on other predictors. In multiple regression |β| can exceed 1; do not apply correlation cutoffs automatically. Type II error β instead lies between 0 and 1. UCLA: regression coefficients	A standardised β = .30 for study hours means a one-SD difference in hours corresponds to .30 SD in quiz scores, holding the other included predictors constant.
Intercept	The model's expected outcome, or linear predictor, when numeric predictors equal zero and categorical predictors are at their reference levels. Its interpretation depends on coding and centring. Also written as: intercept	**Can be negative, zero or positive on the model scale** 0 is the model prediction at the coded zero/reference conditions, not a goodness threshold. An intercept on a logit scale needs transformation to obtain probability.	In predicted score = 50 + 2 × study hours, the intercept is 50: the model's score at zero study hours, if that situation is meaningful.
Slope	The modelled change in the outcome or link scale per unit of a predictor. With an interaction, the slope can depend on another variable; with growth models it often represents change per time unit. Also written as: slope · slopes	**Usually any real value** Positive = increasing modelled outcome/link value; negative = decreasing. A slope of 0 is the no-change/no-association reference, conditional on the specified model.	On a graph of study hours and predicted score, a slope of 2 means the prediction rises 2 points for each extra hour.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
X: predictor variable	X names a predictor in these equations. A one-unit X difference is interpreted on X's stated scale; coding matters when X represents a group. X is a variable label rather than a test statistic. Also written as: X	**Variable label: no inherent numerical range** The range belongs to X's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.	If you use study hours to predict quiz scores, X can stand for study hours. X labels the input variable; it is not a result of the test.
Y: outcome variable	Y names the outcome or response being modelled. In a repeated-measures equation, a subscript identifies the occasion or person. Y is observed; ŷ denotes a model prediction. Also written as: Y	**Variable label: no inherent numerical range** The range belongs to Y's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.	In that study-hours model, Y is each student's actual quiz score: the outcome you are trying to describe or predict.
Z: context-dependent symbol	Z can name an additional predictor, a standardised score or a normal-reference statistic. Read the equation or output label before assigning it a meaning. Also written as: Z	**Variable label: no inherent numerical range** The range belongs to Z's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.	To compare study hours within your model, you might also include prior quiz score as Z. Here Z names another predictor, not a z statistic.
IV: independent/ predictor variable	A variable used to explain or predict an outcome in the model. Calling it independent does not mean it is statistically independent of other predictors or experimentally manipulated. Also written as: IV · independent variable · predictor variable	**Variable label: no inherent numerical range** The range belongs to the predictor's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.	In a model asking whether study hours relate to scores, study hours play the predictor/independent-variable role. That name alone does not show a causal effect.
DV: dependent/ outcome variable	The response being described, compared or predicted. Its type-continuous, binary, ordinal, count or time-to-event-is central to choosing a model. "Dependent variable" is a role label, not proof of causal dependence. Also written as: DV · dependent variable · outcome variable	**Variable label: no inherent numerical range** The range belongs to the outcome's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.	In the same question, quiz score is the dependent/outcome variable: the quantity the model tries to account for.
ŷ: predicted outcome	The model's fitted or predicted value of Y at the stated predictor values. The hat distinguishes a prediction from an observed outcome; an individual observation can differ from this value. Also written as: ŷ	**Follows the fitted model's output scale** A linear prediction may be any real value; a predicted probability must be 0-1, and a predicted count non-negative. Out-of-range linear predictions can flag a mismatch to the outcome.	Your model predicts 76 points for a student with certain study hours. ŷ = 76 is the prediction; the student may actually score 80.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Residual	For an observed continuous outcome, the observed value minus its model-fitted value. Residual patterns help diagnose model problems; a residual is different from a regression coefficient. Also written as: residual · residuals	**Can be negative, zero or positive** Observed − predicted: +2 means underprediction by two units; −2 means overprediction. 0 means an exact fitted value for that observation; raw residuals have no universal ±2 cutoff.	The model predicts 76 points but the student scores 80. The residual is +4 points: actual minus predicted.
Reference category / group	The category against which coded coefficients are compared. If group A is the reference and b for B is 3, the modelled B−A contrast is three units on the stated response/link scale. Also written as: reference category · reference group	**A category/coding choice, not a statistic** A dummy-coded reference is often coded 0, while the comparison is 1. These numbers are labels, not evidence that one group is better.	Compare bus, car and walking journeys using bus as the reference. A car coefficient of −5 means a modelled car journey is 5 minutes shorter than a bus journey.
Covariate	An additional variable included in a model to specify a conditional comparison or relationship. Adjustment does not automatically remove confounding or establish causality. Also written as: covariate · covariates	**Range follows the covariate's measurement** Age may be non-negative; a centred score may be negative. No numerical cutoff determines whether a variable should be adjusted for; the research design and target comparison do.	When comparing two courses' final scores, include starting score as a covariate. The model compares courses at the same starting score; this does not guarantee a causal comparison.
Centring	Subtract a chosen reference, often the sample mean, from a predictor. This changes what zero represents and can make intercepts and interaction main terms easier to interpret. It does not automatically solve all collinearity problems. Also written as: centring · centering	**Centred values can have either sign** Mean-centred 0 means the sample mean, not absence of the trait. +2/−2 means two original units above/below the chosen reference.	Your group studies 5 hours a week on average. Subtract 5 from everyone's hours: zero now means an average amount of study, rather than no study.
VIF: variance inflation factor	A diagnostic of predictor collinearity in a specified regression design. $VIF_j=1/(1-R^2_j)$, where R^2_j comes from regressing predictor j on the others. Interpret it alongside model purpose and design; a single cutoff is not a complete diagnosis. Also written as: VIF · collinearity	**VIF ≥ 1 for the usual intercept-based calculation** 1 means no linear prediction by the other predictors. 5 or 10 are common warning heuristics, not automatic deletion rules. Interpret precision, sample size and the specified design. O'Brien: VIF cautions	A model includes study hours and study minutes. They carry the same information, so separate coefficients cannot be estimated. VIF warns about this overlap; with exact duplication it is unbounded.

Data types and observation structure

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Continuous outcome	An outcome treated as numerical on a scale where differences in magnitude are meaningful. Choosing a linear model still requires examining its form, residual behaviour and dependence. Also written as: continuous outcome · continuous outcomes	**Type or structure: no universal numeric range** Values follow the measurement scale; 0 is meaningful only if defined by that scale.	Journey time measured in minutes is numerical: a 10-minute difference has a meaningful size. This differs from labels such as bus or walking.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Categorical outcome	An outcome taking distinct categories. Binary has two categories; ordinal has a meaningful order; nominal categories have no assumed order. The response type affects the appropriate model. Also written as: categorical outcome · categorical outcomes	**Type or structure: no universal numeric range** Binary labels are often 0/1. Ordered category labels such as 1-5 do not automatically have equal spacing.	Your preferred transport is bus, car or walking. These are categories; coding them 1, 2 and 3 does not make car “twice” bus.
Independent observations	Under the specified model, information about one observation does not change the distribution of another. Different people can still share a cluster. Independence is a property justified by the sampling/design and model, not by having different names. Also written as: independent observations · independent groups	**Type or structure: no universal numeric range** Independence is a design/model property, not established by a particular correlation p-value.	One journey from each separately sampled commuter may be treated as independent when the design supports it. Ten journeys from one commuter are linked instead.
Paired / matched observations	Two linked measurements, such as the same person before and after, or deliberately matched units. A paired t-test evaluates the mean of the within-pair differences and uses their SD and SE. Also written as: paired · matched pairs	**Type or structure: no universal numeric range** Each pair links two measurements. Pair identifiers are labels; analyse within-pair differences using their scale.	Weigh the same person before and after a programme. Keep their two measurements together and analyse their change, rather than mixing everyone’s before and after values.
Repeated observations	Multiple measurements come from the same unit, such as one student assessed three times. Preserve that linkage in the analysis rather than treating each row as an unrelated person. Also written as: repeated observations · repeated measurements	**Type or structure: no universal numeric range** The number and spacing of occasions must be stated. Three scores from one person are not three independent people.	Record your sleep every night for a week. Those seven records come from one person; they are not seven independent people.
Nested observations	Each lower-level unit belongs to one higher-level unit in the design: for example, students within classes. Shared class conditions can induce dependence. A model may need to distinguish within-class and between-class variation. Also written as: nested observations · nested	**Type or structure: no universal numeric range** Cluster labels have no numerical ordering; represent the lower/higher-level structure explicitly.	Students belong to classes, each with its own teacher. Pupils in the same class share conditions; the data have a student-within-class structure.
Crossed observations	Two grouping factors overlap instead of one fitting entirely inside the other. For example, several raters score several essays: ratings can share a rater, an essay, or both. Also written as: crossed observations · crossed	**Type or structure: no universal numeric range** Rater and item identifiers are labels, not magnitudes. An observation may share either grouping factor.	Three teachers each mark the same ten essays. Scores share both teachers and essays; neither grouping fits entirely inside the other.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
k: number or index	k often denotes a number of groups, items, clusters or parameters, or indexes one component. Its definition is local to the model. It is not a universal test statistic. Also written as: k · K	**Usually a non-negative/positive integer or index** k = 3 may mean three groups, items or parameters; it has no fixed meaning or cutoff across models. State the local definition.	For a four-question sleep scale, k = 4 may count items. For a three-group comparison, k = 3 may count groups. Read what k counts in that formula.

Use when the analysis calls for it

Further test terminology: tails, errors and power

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
One-sided test	Tests a prespecified directional alternative. The direction should follow the research question and plan, rather than being chosen after seeing which direction produces a smaller p-value. Also written as: one-sided · one-tailed	**A test choice, not a statistic** A 5% upper-tail normal test uses $z > 1.645$; a lower-tail test uses $z < -1.645$. The direction must be specified before examining the result. NIST: normal critical values	Before collecting data, you specifically plan to test whether a new queue system reduces waiting. A one-sided test targets that pre-agreed direction.
Two-sided test	Allows departures from the null in either direction. In a simple symmetric normal-reference test with α=.05, the critical values are approximately −1.96 and +1.96. Also written as: two-sided · two-tailed	**A test choice, not a statistic** At α = .05 a standard-normal two-sided test allocates .025 to each tail; critical values are approximately ±1.96. Other reference distributions differ. NIST: normal critical values	You ask whether a new queue system changes waiting time in either direction. A two-sided test allows both shorter and longer waits.
False positive / Type I error	A false positive in hypothesis testing means rejecting a null hypothesis that is actually true. For example, a test reports a significant mean difference even though the population means being compared are equal. Repeating many tests without adjustment increases the chance of at least one false positive. A prespecified significance level such as .05 controls the long-run error rate for a single test under its assumptions; it does not mean a significant result has a 5% probability of being wrong. Also written as: false positive · false positives · Type I error	**An error event; its probability is 0-1** A prespecified α = .05 targets a 5% long-run false-rejection rate under the null and the procedure's assumptions. It is not a 5% probability that this conclusion is wrong. ASA: interpreting p-values	Two queue systems truly have the same population mean wait, but your sample leads you to declare a significant difference. That is a false positive.
β: Type II error probability	Failing to reject H_0 under a specified alternative. This β depends on effect size, sample size, variability and the test; it is a different meaning of β from a standardised regression coefficient. Also written as: Type II error	**$0 \le \beta \le 1$** β = .20 means a 20% chance of not rejecting H_0 under the specified alternative; power is then .80. This is not the regression coefficient β.	A new queue system really reduces the population mean wait, but your test does not detect it in this sample. That is a missed effect: a Type II error.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
Power = 1−β	The probability that a specified test rejects H_0 under a specified alternative. Power=.80 means 80% in that scenario. Power is not the probability that a non-significant result proves no effect. Also written as: power · 1−β	**0 ≤ power ≤ 1** .80 and .90 are common planning targets: 80% or 90% rejection probability under the specified alternative. They require a stated effect, α, design and sample size. Cohen: conventional effect sizes	Imagine repeating a study of a specified real 3-minute queue reduction. Power = .80 means the test detects it in about 80% of those repetitions under the assumptions.

Probabilities, odds, counts and event timing

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
P / p: event or response probability	In probability equations P or p denotes a probability between 0 and 1. In logistic regression p=.426 means a 42.6% modelled event probability; it is not a hypothesis-test p-value. P(A|B) means the probability of A conditional on B. Also written as: P · p · event probability · predicted probability	**0 ≤ P ≤ 1** 0 = impossible under the model; 1 = certain; .50 = 50%. A predicted probability of .05 is not a p-value or a significance decision.	If your model gives a .70 probability of taking the bus, it predicts a 70% chance for that outcome, not a hypothesis-test p-value.
OR: odds ratio	A ratio of odds, not probabilities. In a logistic model, exp(b) gives the odds ratio per one-unit predictor difference with other included predictors held constant. OR=1 is the no-association reference; OR=2 does not generally mean twice the probability. Also written as: odds ratio · odds ratios · OR	**OR > 0 for finite log-odds contrasts** 1 = equal odds; 2 = twice the odds; .50 = half the odds. These are not probability ratios. For inference, compare a corresponding interval with 1.	If 2 in 3 bus users and 1 in 2 walkers arrive on time, their odds are 2:1 and 1:1. The odds ratio is 2, although the probabilities are not doubled.
Logit	logit(p)=ln[p/(1−p)], the log of the odds. Logistic regression models this scale, so a coefficient on the logit scale is not directly a probability difference. Also written as: logit	**−∞ to +∞ for 0 < p < 1** logit = 0 corresponds to p = .50; +1 to about .731; −1 to about .269. Values approach infinities as p approaches 1 or 0.	A 50% chance of taking the bus has odds 1:1 and logit = 0. Logistic regression works on this transformed scale before converting predictions back to probabilities.
exp: exponential function	exp(x)=e^x. Exponentiating a logistic-regression coefficient gives an odds ratio; exponentiating a count-model log-rate coefficient gives a rate ratio. Also written as: exp	**exp(x) > 0 for finite x** exp(0) = 1; exp(.693) ≈ 2. A negative log coefficient produces a ratio below 1, not a negative ratio.	A logistic model's bus-use coefficient is about .69. Applying exp gives about 2, turning the coefficient into an odds ratio that is easier to interpret.
IRR: incidence-rate ratio	For a count model with a log link, exp(b) expresses a multiplicative change in the expected count, or rate when exposure is appropriately incorporated. IRR=1.35 means a 35% higher modelled count/rate, conditional on included predictors. Also written as: IRR · incidence-rate ratio	**IRR > 0 for finite log-rate contrasts** 1 = equal modelled counts/rates; 1.35 = 35% higher; .70 = 30% lower. For rates, exposure must be included appropriately; interval reference is 1.	A travel model gives IRR = 1.5 for route A versus B. With equal observation time and other predictors held constant, A's expected delay count is 50% higher.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
HR: hazard ratio	A ratio of event hazards under the specified survival model, conditional on being event-free up to that time. HR=1.5 does not directly mean the cumulative probability of the event is 50% higher. Also written as: HR · hazard ratio	**HR > 0 for finite log-hazard contrasts** 1 = equal hazards; 1.50 = 50% higher hazard among those still event-free. It does not mean 50% higher cumulative event probability; interval reference is 1.	In a study of time until someone stops using an app, HR = 1.5 compares the current quitting rates among people still using it; it does not mean 50% more users eventually quit.

Measurement and model fit

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
α: alpha	In reliability analysis, α is Cronbach's internal-consistency coefficient. In hypothesis testing, α is a prespecified Type I error threshold, such as .05. These are different uses of the same symbol. Also written as: α · alpha	**Usually 0-1 in a coherent scale; estimates can be negative** For reliability alpha, .70 is a common rough screening benchmark, not universal adequacy. Very high values (around .95+) can suggest redundant items. Negative alpha calls for checking scoring and item covariance. Testing α has a different meaning. Tavakol & Dennick: alpha · UCLA: negative reliability estimates	A five-question sleep scale asks about related sleep problems. Reliability α summarises how the answers hang together; a high value alone does not show that sleep is measured validly.
λ: factor loading	In a factor model, λ links an indicator to a latent factor. Its scale depends on standardisation and identification. A standardised loading .80 implies .64 explained variance only under the stated simple-model assumptions. Also written as: λ · factor loading · factor loadings · loading · loadings	**Scale/model dependent; loadings may have either sign** In a simple standardised one-factor model, $\|\lambda\| \leq 1$ and λ = .70 implies .49 explained variance. Unstandardised or cross-loading coefficients need not share that bound. A simple-model loading above 1 can indicate an improper solution.	A hidden sleep-quality factor is linked to an item such as "I wake feeling rested". Its loading λ describes how strongly that item reflects the factor in the model.
θ: latent trait or threshold	In an item-response model, θ usually denotes a person's latent trait level. In an ordinal regression equation, θ_j may denote category threshold j. The equation determines the interpretation. Also written as: θ	**Usually any real value on the latent/link scale** An IRT scale is often identified with mean 0 and SD 1; θ = +1 is then one latent SD above the reference mean, not a 1% gain. Ordered thresholds must follow their specified ordering.	In an item-response model of a vocabulary quiz, θ represents a learner's underlying vocabulary ability. It is an estimated ability level, not simply the number correct.
h^2: communality	The part of an indicator's variance accounted for by the common factors under the fitted factor model. The sum of squared loadings is sufficient only under the stated orthogonal-factor conditions. Also written as: h^2	**$0 \leq h^2 \leq 1$ for an admissible standardised factor model** h^2 = .64 means common factors account for 64% of that indicator's variance. Values above 1 are a warning of an inadmissible/Heywood-type solution, not especially strong measurement.	A sleep questionnaire item varies across people. Its communality is the share of that variation accounted for by the model's common factors.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
φ: context-dependent association	For a 2×2 table, φ is an association effect size related to chi-square. In factor-model notation, φ may instead denote a factor correlation or covariance. Read the model context. Also written as: φ	**−1 to +1 for correlations; covariance has other units** For a 2×2 association or factor correlation, 0 indicates no association on that metric. If φ denotes covariance, its numerical range depends on the variables and is not fixed at ±1.	A factor model may use φ for the correlation between sleep quality and stress. A 2×2 bus-use/on-time table may use φ for a different association measure; check the setting.
ICC: intraclass correlation	In a simple Gaussian random-intercept model, ICC=between-cluster variance/(between-cluster variance+residual variance). ICC=.20 means 20% of the modelled variance is between clusters. Other ICC definitions serve other designs. Also written as: ICC · intraclass correlation	**0-1 for the simple variance-component definition** ICC = .20 means 20% of modelled variance is between clusters. Other sample ICC estimators can be negative. There is no universal cutoff for ignoring clustering or declaring rater agreement adequate.	In a simple school model, ICC = .20 means 20% of score variance lies between classes. It helps explain why classmates cannot always be treated as unrelated observations.
CFI: comparative fit index	An incremental fit index comparing a specified factor/SEM model with a baseline model. Read it alongside residuals, other fit indices, uncertainty and model plausibility; a single cutoff does not validate a model. Also written as: CFI	**Usually 0-1 in bounded implementations** Closer to 1 indicates better relative fit. CFI near .95 is a widely cited guideline from specific ML simulations, not proof of validity or a universal rule for ordinal/other estimators. Hu & Bentler: fit-index guidelines · Xia & Yang: estimator-dependent fit	You propose a model linking sleep-questionnaire items to sleep quality. CFI checks its improvement over a simpler baseline model; it is not the percentage of people correctly classified.
TLI: Tucker-Lewis index	An SEM incremental fit index comparing model fit with a baseline while accounting for model degrees of freedom. Interpret it with the model specification, other fit indices and residual evidence. Also written as: TLI	**Often near 0-1; may fall below 0 or exceed 1** Closer to 1 generally indicates better relative fit. Near .95 is a commonly cited guideline in particular ML conditions. TLI = 1.02 is possible; read the estimator, df and other diagnostics. Hu & Bentler: fit-index guidelines · Xia & Yang: estimator-dependent fit	For the same sleep questionnaire, TLI compares fit with a baseline while considering model complexity through df. Adding more paths does not automatically make the model better.
RMSEA	Root mean square error of approximation: an SEM fit index addressing approximate discrepancy relative to degrees of freedom. Read its interval and the model's df; small values alone do not establish a sound model. Also written as: RMSEA	**RMSEA ≥ 0; no universal upper bound** Lower indicates less approximate misfit; around .06 or below is a cited ML guideline, not an automatic pass. Read its interval and df; values can behave poorly at low df or small samples. Hu & Bentler: fit-index guidelines · Xia & Yang: estimator-dependent fit	Your sleep-questionnaire model cannot perfectly reproduce the relationships between answers. RMSEA summarises approximate mismatch while accounting for df; read it with its interval.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
SRMR	Standardised root mean square residual: a summary of standardised discrepancies between observed and model-implied association structures. A small overall index can still conceal important local misfit. Also written as: SRMR	**SRMR ≥ 0** 0 means no discrepancy in the summarised standardised residuals; around .08 or below is a cited fit guideline. A small average can hide local misfit; read jointly with other diagnostics. Hu & Bentler: fit-index guidelines	Compare the actual relationships between sleep-questionnaire items with those predicted by the model. SRMR summarises the standardised gaps; a small average can still hide a poorly fitted pair.
BIC: Bayesian information criterion	A relative model-comparison criterion balancing fit and parameter count. Lower BIC is preferred among comparable candidate models fitted to the same data. It is not a p-value or an absolute certificate of model quality. Also written as: BIC	**May be negative or positive** Lower is preferred among appropriately comparable models fitted to the same data. BIC = −20 can be better than −10. Zero is not a pass/fail reference; neither the sign nor BIC alone selects a defensible model.	Compare two- and three-profile models of students' study habits on the same data. BIC balances fit against extra parameters; a lower value favours one candidate, not a proven truth.
AIC: Akaike information criterion	A relative model-comparison criterion balancing fit and complexity. Lower is preferred among appropriately comparable models using the same data. AIC and BIC impose different complexity penalties. Also written as: AIC	**May be negative or positive** Lower is preferred among appropriately comparable models using the same response/data and likelihood basis. AIC = 100 versus 104 favours 100 on this criterion; no absolute "good AIC" exists.	Compare two models predicting the same students' quiz scores from the same data. AIC weighs fit against complexity; the model with more predictors need not win.

Mediation, moderation and latent groups

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
M: mediator	In mediation notation, M is the intervening variable in the proposed X→M→Y pathway. Calling it a mediator in an equation does not by itself establish a causal mechanism. Also written as: M	**Variable label: no inherent numerical range** The range belongs to the mediator's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.	You propose that study hours (X) relate to confidence (M), which relates to quiz score (Y). M names the proposed middle step; the data alone do not prove this mechanism.
W: moderator	In the interaction examples, W is the moderator. The X-Y slope is evaluated at stated values of W. Different coding or centring changes which conditional slope is the main coefficient. Also written as: W	**Variable label: no inherent numerical range** The range belongs to the moderator's measurement or coding: for example 0/1 groups, a 1-5 score or unrestricted values. The letter itself has no cutoff.	The link between coffee and alertness may differ with sleep hours. Sleep is the moderator W: it tells you when the coffee-alertness relationship is stronger or weaker.
a: context-dependent coefficient	In mediation, a labels the X→M path coefficient. In a 2PL item-response model, a is item discrimination, describing the steepness of the response curve. Read the model before interpreting the symbol. Also written as: a	**Usually signed; interpretation depends on the model** A mediation path a can have either sign, with 0 as the no-association reference. IRT discrimination is commonly positive for correctly keyed items; constraints and direction conventions must be checked.	In the proposed study-hours → confidence → quiz-score pathway, a labels the first link: study hours to confidence.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
ab: indirect component	For the specified linear mediation model without an X×M interaction, multiply a (X→M) by b (M→Y conditional on X). This gives the indirect component ab; its uncertainty needs an appropriate interval method. Also written as: ab	**Signed; no universal numerical bounds** The indirect product can be positive or negative. In the specified linear model, an appropriate interval excluding 0 supports a nonzero indirect component; it does not establish a causal mechanism.	If the study-hours → confidence coefficient is .5 and confidence → score is 2, their product is 1. In this linear model, ab = 1 is the indirect component.
c′: direct path coefficient	In the stated mediation model, c′ labels the X→Y coefficient conditional on mediator M and the other included variables. This statistical direct path requires additional assumptions for a causal interpretation. Also written as: c′	**Signed; no universal numerical bounds** 0 is the no-direct-association reference on the specified coefficient scale, conditional on M and other included variables. A nonsignificant c′ does not establish “full mediation”.	In that pathway model, c′ is the study-hours → score coefficient after including confidence and the other stated predictors. Calling it “direct” does not itself prove causation.
π: mixture proportion	In a mixture model, π_k is the modelled proportion of cases in profile or class k. Proportions across all components sum to 1. A class proportion differs from an individual’s posterior membership probability. Also written as: π_k · π_1 · π_2 · π	**0 ≤ πk ≤ 1; class proportions sum to 1** πk = .30 means a modelled 30% class/profile proportion. It is not the probability that one particular learner belongs to that class; no fixed minimum class proportion validates a solution.	A fitted study-habit model estimates 30% of students in profile 1. π_1 = .30 is the group’s modelled proportion, not one student’s membership probability.
Entropy in latent grouping	A summary of posterior classification separation for one fitted class/profile solution. Higher separation does not by itself determine the number of groups or show that groups are natural types. Also written as: entropy · Entropy	**0-1 for the normalised LPA/LCA index** Closer to 1 means clearer classification; .80 is often used as a rough guide to separation. It is not a p-value, a percentage correctly classified, or a rule for choosing the class count. Other entropy definitions have different ranges. Mplus: entropy and classification	A study-habit model assigns most students clearly to one profile rather than being unsure between several. Higher entropy describes that clarity, not whether the chosen profiles are correct.

Estimation, prediction and research synthesis

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
OLS: ordinary least squares	Fits a linear model by minimising the sum of squared residuals. The estimator, inferential standard errors and independence assumptions are separate choices that should be specified. Also written as: OLS	**An estimation method, not a bounded statistic** OLS does not make all coefficients positive or impose a significance cutoff. Coefficient ranges follow the specified model.	Draw a line predicting commute time from distance. OLS chooses the line with the smallest total squared vertical gaps between actual times and predictions.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
ML: maximum likelihood	Chooses parameters that maximise the probability density/mass of the observed data under the specified model. ML is an estimation principle, rather than a substantive analysis method or guarantee that the model is correct. Also written as: ML · maximum likelihood	**An estimation method, not a bounded statistic** A likelihood is non-negative, but a likelihood density can exceed 1 and a log likelihood can have either sign. ML itself is a fitting principle, not a number to compare with .05.	Imagine trying several values for the probability of taking the bus. ML chooses the value that makes the observed pattern of bus/non-bus choices most likely under the model.
FIML: full-information ML	Uses each case's observed measurements in a model-based likelihood for incomplete data. Validity relies on the model and missingness assumptions, such as MAR with appropriate observed variables. It does not mean every missing value is known or filled in. Also written as: FIML	**An estimation method, not a bounded statistic** FIML has no numerical pass mark and is not the percentage of complete data. Interpret the resulting parameters and the missing-data assumptions.	A student answers the sleep items but misses a stress item. FIML can use their available answers under its model and missing-data assumptions instead of discarding the whole record.
Bootstrap	Repeatedly resamples the data under a justified scheme and recomputes a statistic to approximate its sampling uncertainty. Resample people or clusters when the design requires it; resampling individual rows may break dependence. Also written as: bootstrap · resampling	**An estimation method, not a bounded statistic** The number of replicates is a positive integer, but there is no universal bootstrap cutoff. The statistic's range and the interval type determine interpretation.	You have 50 commuters. Draw a new sample of 50 from them with replacement, calculate the mean, and repeat. The spread of those means helps estimate uncertainty.
Posterior probability	A probability updated using data under a specified model. In LPA/LCA it can express a case's uncertain profile/class membership. A highest-probability assignment does not make membership certain. Also written as: posterior probabilities · posterior probability	**0-1 for a posterior probability** A class probability of .80 means 80% membership probability under the fitted model, not certainty. Posterior densities, unlike probabilities, may exceed 1; posterior parameters have their own domains.	A study-habit model gives one student .80 probability of profile A and .20 of B after using their answers. A is more likely, but membership remains uncertain.
Credible interval	A Bayesian posterior interval containing a stated probability mass under the model and prior. Its probability interpretation differs from the repeated-sampling coverage of a frequentist confidence interval. Also written as: credible interval · credible intervals	**Endpoints follow the parameter domain** A 95% credible interval contains 95% posterior probability under the model and prior. Compare signed effects with 0 and ratios with 1, but do not equate posterior coverage with frequentist confidence coverage.	A Bayesian model gives a 95% credible interval of [28, 32] minutes for mean commute time. Given its prior, model and data, that interval contains 95% of the posterior probability.
Prediction interval	An interval for a future observation or effect under a specified model. It includes relevant future variability, and answers a different question from a CI for a mean or coefficient. Also written as: prediction interval · prediction intervals	**Endpoints follow the predicted outcome/effect** A 95% prediction interval targets future variation under the specified model. It can span 0 even when the CI for the mean effect does not; there is no universal acceptable width.	A mean-commute CI may be [28, 32] minutes, while a prediction interval for one new commuter is [10, 50]. Individual journeys vary more than an estimated mean.

Term / notation	Meaning	Range and numerical interpretation	Everyday example: in plain language
RMSE: prediction error	Root mean squared prediction error: square root of the mean of (observed−predicted)². It is in outcome units; lower is better on the same evaluation data and task. Training RMSE is not an independent validation result. Also written as: RMSE	**RMSE ≥ 0** 0 = exact predictions on the evaluated data. Lower is better for the same task and evaluation data; RMSE = 2 is two outcome units of root-mean-square error, not 2%.	An app predicts journey times. RMSE describes prediction error in minutes, with large misses weighing more heavily; compare apps using the same journeys.
τ^2: between-study variance	In a random-effects meta-analysis, τ^2 describes heterogeneity in the underlying study effects, beyond their sampling variances. It is a variance, expressed in squared effect-size units. Also written as: τ^2	**$\tau^2 \geq 0$ in a constrained random-effects model** 0 indicates an estimated absence of between-study variance, not proof of homogeneous true effects. $\tau^2 = .04$ corresponds to $\tau = .20$ in the effect-size units; no universal small/large cutoff applies.	Several schools study the same tutoring programme. τ^2 describes how much their underlying programme effects vary beyond sampling noise; it is not variation between individual pupils.

Make quantitative analysis and results transparent

Alongside JARS-Qual in Topic 7, use JARS-Quant for this lesson. This classroom companion condenses reporting prompts; consult the complete table and target journal. A tick means you can locate the reported evidence, not that the analysis is valid.

APA JARS-Quant · Table 1

[ ] **Questions, hypotheses and status**Name the target quantities; distinguish primary, secondary and exploratory analyses.

[ ] **Design, sampling and setting**Describe recruitment, inclusion/exclusion, selection, settings and dates.

[ ] **Sample size, power and precision**Explain intended and achieved sample sizes and their rationale; report participant flow.

[ ] **Ethics and data collection**Identify ethical review, consent and procedures relevant to the study.

[ ] **Measures and scoring**Define variables, covariates, units, item coding and composite-score rules.

[ ] **Measurement evidence**Report relevant reliability in the current sample and validity/comparability evidence.

[ ] **Missingness and exclusions**Report amounts/patterns, handling, outlier decisions, exclusions and final analysis n.

[ ] **Model, estimand and estimation**Specify variables, contrasts, estimators, dependence, weighting and transformations.

[ ] **Assumptions and diagnostics**Explain relevant checks, estimation failures and responses to violations.

[ ] **Descriptives and all relevant results**Provide sample summaries and planned results, including non-significant findings.

[ ] **Magnitude and uncertainty**Report estimates, units, intervals, effect sizes and inferential statistics as applicable.

[ ] **Multiplicity and sensitivity**Disclose multiple tests, adjustments, alternative models and unplanned changes.

[ ] **Tables, figures and reproducibility**Label units, n, intervals and notes; identify software, versions and access to data/materials/code where appropriate.

[ ] **Interpretation and limitations**Discuss precision, generalisability, alternatives and design-based limits on causal claims.

Thank you

Thank you for your questions and participation. Start with the research question, show the data clearly and explain the uncertainty.

Download the English lesson PDF to revisit the foundations, four detailed methods, paper-reading questions, glossary and JARS-Quant checklist. Each study PDF is available beside its reading activity. The method map remains available for independent exploration.

DEDC02 course materials